Computer Network Note

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Contents at a Glance:
This note covers the following major topics:

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Protocols, Peers, Network Architecture, Protocol Stack
Design Issues for the Layers (4)

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Book: Computer Networks by Tanenbaum 5th Edition

Wk1-1-Inroduction

Computer Network is the collection of autonomous computers interconnected by a single technology.

Two computers are said to be interconnected if they are able to exchange information. The connection need not be via a copper wire; fiber optics, microwaves, infrared, and communication satellites can also be used.

A distributed system is a setup where many independent computers are connected through a network and work together to complete tasks. They share the workload among themselves and coordinate with each other so that, to the user, the whole system appears as one single system rather than many separate computers.

Computer Network vs Distributed System

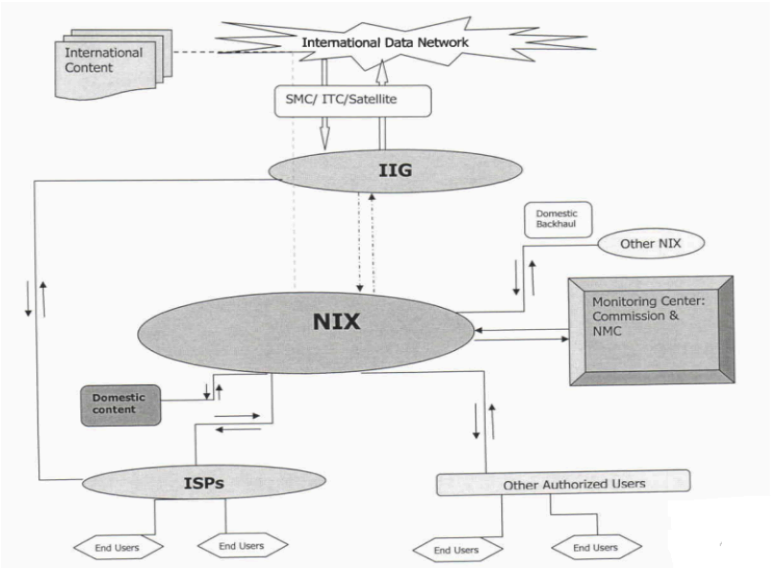
1. In a distributed system, many computers work together but look like one single computer to the user. A software layer called middleware hides the differences between the machines and makes them act as a unified system. It is built on top of a network.

2. In a computer network, computers are connected to each other, but each one is separate. Users can see and access each machine individually and must connect to them manually when needed.

So, the main distinction lies in the software layer, not the hardware.

International Internet Gateways (IIGs)
National Internet Exchanges (NIXs)

ISO (International Organization for Standardization):
An international organization that creates global standards for technology, quality, and safety.



Applications of Computer Networks

1) Business Applications
2) Home Applications
3) Mobile Users
4) Social Issues

Business Applications

1. Primary Goal: Resource Sharing

Share hardware: printers, scanners, storage
More important: sharing data (customer records, files, databases)

Example:

1) VPN (Virtual Private Network): secure remote access
2) Client–Server Model: servers provide services; clients request (e.g., web applications)

2. Secondary Goal: Communication Medium

1. Email
2. VoIP
3. Video calling / conferencing
4. Desktop sharing: TeamViewer, AnyDesk

3. Tertiary Goal: E-Commerce

1) Electronic commerce (online buying/selling)

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books online
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products online
P2P	Peer-to-peer	Music sharing

Figure 1-4. Some forms of e-commerce.

Home Applications

1. Connectivity (home users can)

- ✔ Connect to remote computers
- ✔ Access information from anywhere
- ✔ Communicate with others through messaging, email, or calls
- ✔ Buy products and services online (e-commerce)

2. General Uses

- ✔ Surf the internet for entertainment such as movies, videos, and chatting
- ✔ Read news from online news portals
- ✔ Perform academic research using platforms like IEEE and Springer for journals and papers

Networking Models Used

Networking models describe how computers interact and share data over a network. The two main models are **Client–Server** and **Peer-to-Peer (P2P)**.

1. Instant messaging
2. Twitter, Facebook, WhatsApp
3. Wikipedia (community-created content)
4. E-commerce from home

Mobile Users

Connectivity through mobile data / Wi-Fi

1. Mobile hotspot
2. Texting / SMS
3. GPS navigation
4. M-Commerce: mobile-based online transactions
5. NFC (Near Field Communication): contactless payments

Social Issues

Networking and the internet bring many benefits, but they also create several social challenges that affect individuals and society.

1) Privacy concerns
2) Security threats / cybercrimes
3) Copyright & digital rights
4) Impact of social media on society
5) Spread of misinformation
6) Ethical use of technology

Transmission Technology

Computer networks can be classified by their transmission technology.
There are two widely used types: Point-to-Point Links and Broadcast Links

1. Point-to-Point Links

Point-to-point links connect two machines for communication, where data is sent as packets either directly or through intermediate nodes.

- ✔ Data travels as packets from one machine to another.
- ✔ Packets may pass through one or more intermediate machines.
- ✔ Multiple routes may exist, so choosing an efficient route is important.
- ✔ **Communication between one sender and one receiver is called unicasting**.

2. Broadcast Links

Broadcast links are communication channels shared by all machines in a network, where a packet sent by one machine is received by all others.

1) Each packet includes an address field to specify the intendedrecipient.
2) When a machine receives a packet:
   i. If the address matches, it processes the packet.
   ii. If not, it ignores the packet.
3) Wireless networks are common examples of broadcast links.

Multicast Links

Some systems allow sending a packet to a selected group of machines only, which is called multicasting.

Classify Network By Scale

Networks can be classified based on their geographic size and coverage area.

Personal Area Network (PAN): A very small network used for personal devices, such as Bluetooth connections between a phone and earbuds.

Local Area Network (LAN): A network that covers a small area like a home, office, or building.

Metropolitan Area Network (MAN): A network that covers a city or large campus; includes technologies like cable networks and WiMAX.

Wide Area Network (WAN): A large network that spans cities, countries, or continents, connecting multiple LANs and MANs together.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Figure 1-6. Classification of interconnected processors by scale.

Network Software

Early computer networks focused mainly on hardware; software was secondary. This approach no longer works due to increasing complexity.
To manage this complexity, networks are organized as a stack of layers, each built on the one below it.

1. Protocols and Network Architecture

Protocol: An agreement between communicating parties on how communication will proceed.
Peers: Entities on the same layer in different hosts that communicate with each other.
Network Architecture: The complete set of layers and protocols in a network.
Protocol Stack: The list of protocols used by a system, with one protocol per layer.

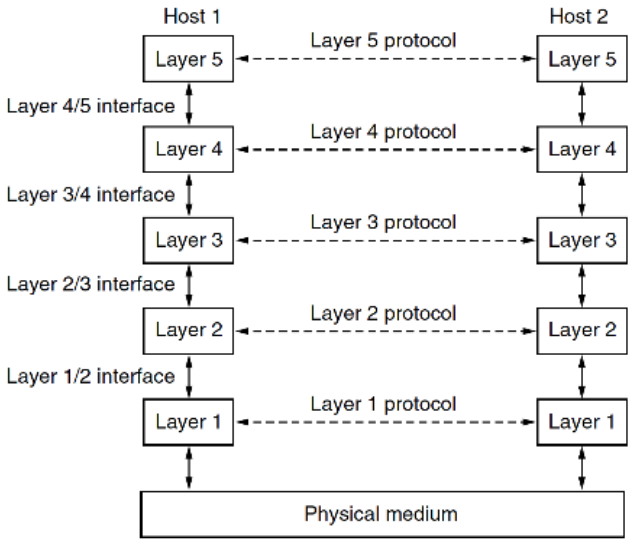
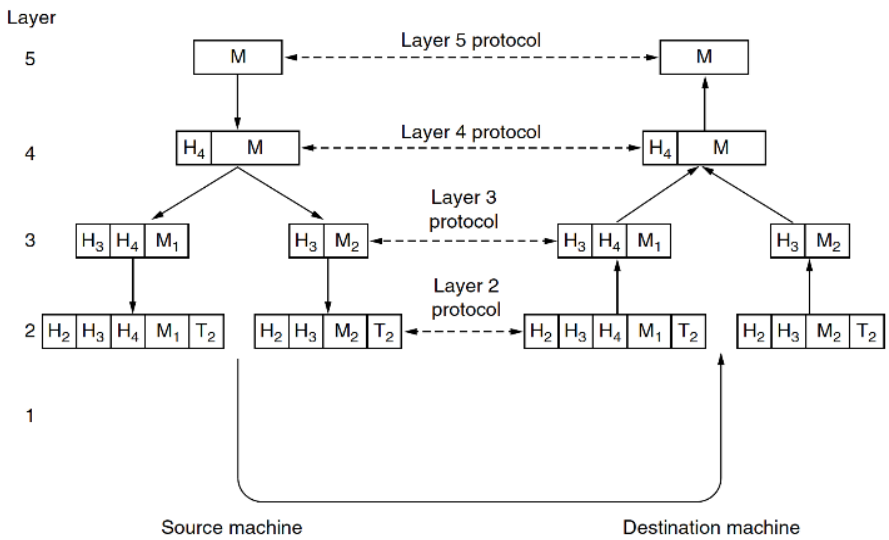


Figure:



2. Design Issues for the Layers

Design issues in network layers focus on how data is transmitted efficiently, reliably, and securely across a network.

Reliability: Ensuring data is delivered correctly using error detection and error correction, mainly handled in lower layers.

Routing: Finding an efficient and working path for data to travel through the network. It also involves addressing/naming and inter-networking (connecting different networks).

Resource Allocation: Managing network resources by deciding who gets bandwidth, using priority management, flow control, and ensuring Quality of Service (QoS).

Security: Protecting data through confidentiality (privacy), authentication (verifying identity), and integrity (ensuring data is not altered).

Reference Models

Reference models define how network communication is organized into layers. Two popular models are OSI and TCP/IP.

OSI (Open Systems Interconnection):
A general layered model developed by ISO. Its protocols are not widely used today, but the model is still important for understanding networking concepts. It consists of 7 layers: Application, Presentation, Session, Transport, Network, Data Link, and Physical. A common mnemonic is "All People Seem To Need Data Processing."

TCP/IP:
A practical model whose protocols are widely used in real-world networks like the internet. The model itself is simpler and less emphasized than its protocols.

OSI Reference Model Layers

The OSI model divides network communication into 7 layers, each with specific functions to ensure proper data transmission.

1. Physical Layer
Transmits raw bits over a communication channel. It deals with electrical signals, bit duration, duplex communication, connection setup/teardown, and connector pins.

2. Data Link Layer
Converts raw transmission into error-free frames. It breaks data into frames, sends them sequentially, and uses acknowledgements for confirmation.

3. Network Layer
Handles routing of packets from source to destination and manages quality of service such as delay, transit time, and jitter.

4. Transport Layer
Accepts data from upper layers, divides it into smaller units if needed, and ensures all parts are delivered correctly to the destination.

5. Session Layer
Establishes and manages sessions between different machines. It provides dialog control, token management, and synchronization.

6. Presentation Layer
Handles data format, syntax, and semantics. It ensures compatibility between different systems and supports encoding and higher-level data structures.

7. Application Layer
Provides network services directly to users through applications such as HTTP for web browsing, file transfer protocols, and email protocols.

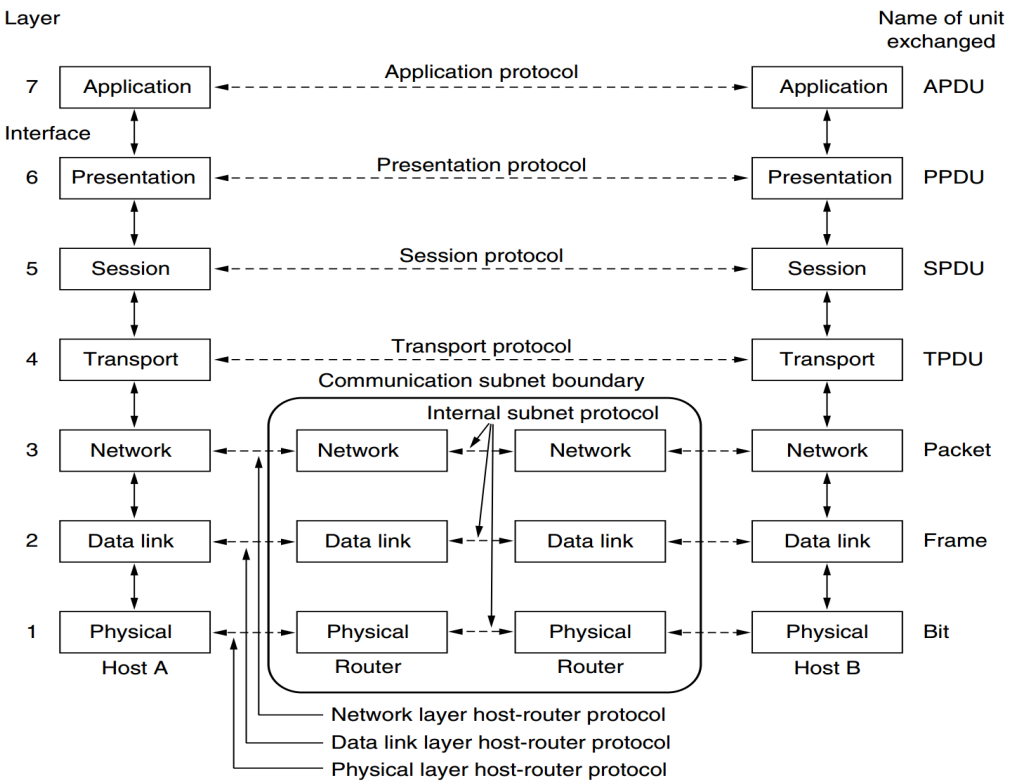


Figure 1-20. The OSI reference model.

TCP/IP Reference Model:

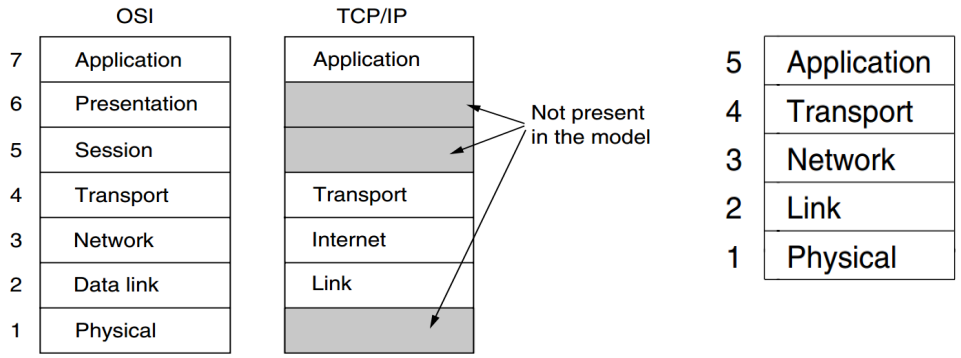


Figure 1-21. The TCP/IP reference model.

OSI Model:

The OSI model is a way to understand how data moves from one computer to another by dividing the process into layers.

Service Definition:
Describes *what a layer does*, not how it works or is accessed.

Layer Interface:
Defines *how upper layers use the service* (inputs and outputs).

Peer Protocols:
Rules for communication between the *same layer* on different machines; can change without affecting services.

Why OSI Model Failed:

David Clark's Theory: Apocalypse of the Two Elephants

David Clark of MIT criticized network standards based on technology, implementation, and politics:

1. **Bad Timing**
   - ✔ OSI was developed too late. By the time it was ready, TCP/IP was already:
   - ✔ Working, Tested. Widely used(in universities and research networks)
   - ✔ So people preferred a working solution over a theoretical one.

2. **Bad Technology**
   - ✔ The seven-layer OSI model was more political than technical.
   - ✔ Session and Presentation layers are almost empty.
   - ✔ Data Link and Network layers are overloaded.

3. **Bad Implementations**
   - ✔ Initial OSI implementations were huge, complex, and slow due to the model's complexity.

4. **Bad Politics**
   - ✔ TCP/IP became popular in academia (UNIX systems).
   - ✔ OSI was promoted by European telecom ministries, the European Community, and the U.S. government, making adoption slower.

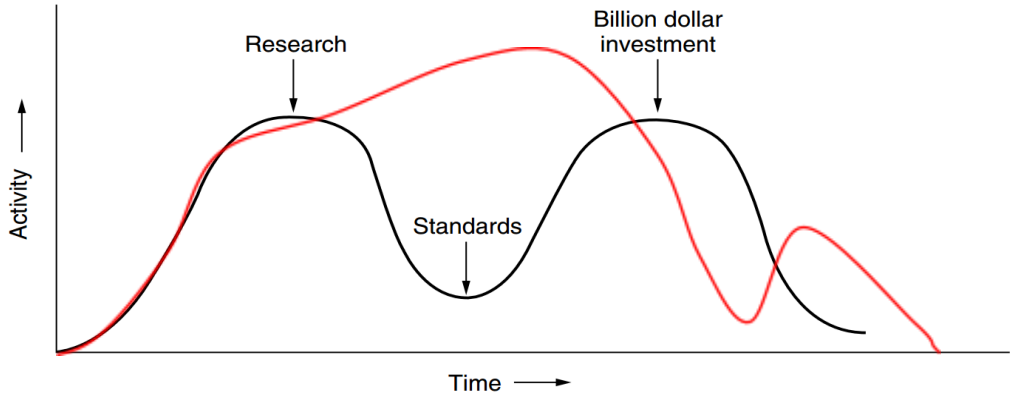


Figure 1-24. The apocalypse of the two elephants.

Network: ARPANET

- ✔ First operational computer network; foundation of the modern Internet.
- ✔ **Origins:** Late 1950s, during the Cold War, the U.S. DoD (Department of Defense) wanted a command-and-control network that could survive even under nuclear attack.

Key Milestones

1960:

1. DoD gave contract to RAND Corporation
2. Paul Baran proposed distributed switching
3. Accepted by Pentagon, rejected by AT&T

1961:

1. U.S. President created ARPA (Advanc**ed Research P**rojects Agency).
2. Funded research via universities & companies not by them

1967:

1. Larry Roberts inspired by Donald Davies
2. Decision to build ARPANET

Network Design

Nodes: IMPs (minicomputers - Interface Message Processor System)
IMPs are small computers (like early routers) that forward data between systems.

Connectivity: Each IMP connects to ≥2 others (reliability)
Each IMP is connected to at least 2 IMPs so that if one path fails, data can take another path.

Speed: 56 kbps
Data transfer speed was 56 kilobits per second, which is very slow compared to today.

Type: Datagram network (automatic rerouting)
Data is divided into packets and each packet can travel independently using different paths.
If a path fails, packets are automatically rerouted.

Software Design

1. Two Parts

Subnet software: IMP communication + reliability
This software runs inside the network (IMPs) and ensures:

1) IMPs can communicate with each other
2) Data is delivered correctly and reliably

Host software: Sends data to IMP, splits into packets
This runs on the user's computer and:

1) Sends message to IMP
2) Breaks large message into small packets

2. Packet Details
Message size: up to 8063 bits. A full message can be this large before splitting.
Packet size: ≤1008 bits. The message is divided into smaller pieces (packets), each with a maximum size.

- ✔ Each packet sent independently and fully received before forwarding
- ✔ Each packet travels separately through the network
- ✔ An IMP first receives the entire packet, then sends it to the next IMP
- ✔ Finally, all packets are reassembled at the destination

"We send data to IMPs, then packets travel through multiple IMPs, and finally reach the destination where they are reassembled."

Wk2-1-Switching-X.25 Protocol

Chapte: 02 (Physical Layer)

X.25 is an early **packet-switching network protocol** used for communication over long distances.

Physical Layer

Concerned with transmitting raw bits over a communication channel.

1. **Design issues include:**
   a) Ensuring 1s and 0s are correctly transmitted
   b) Electrical signals for 1 and 0
   c) Bit duration (nanoseconds)
   d) Duplex communication (simultaneous transmission)
   e) Connection setup and teardown
   f) Network connector pins and their usage

2. **Transmission Medium**

Guided Transmission Medium: Signals transmitted via physical path (e.g., copper wire, optical fiber).
Unguided Transmission Medium: Signals transmitted through the air (e.g., radio, microwave, infrared).

Switching

Definition: Directing a signal or data element to a specific hardware destination.

Two types of switching:

1) Circuit Switching – a dedicated path is established between sender and receiver.
2) Packet Switching – data is divided into packets that are sent independently and can take different paths.

Circuit switching vs Packet Switching

Item	Circuit switched	Packet switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
Time of possible congestion	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Charging	Per minute	Per packet

X.25 Protocol

Developed in the 1970s by ITU-T

(ITU-T = ITU Telecommunication Standardization Sector)

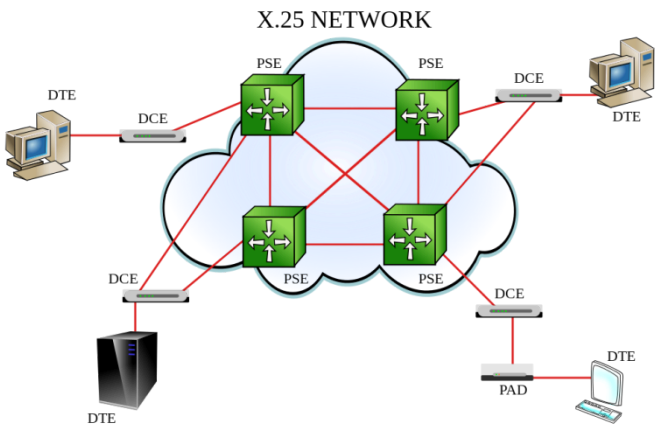
(ITU = International Telecommunication Union)

Main protocols:

1) PLP – Packet Layer Protocol
2) LAP-B – Link Access Protocol-Balanced

X.25 Network Device Categories

1. DTE – Data Terminal Equipment
2. DCE – Data Circuit-Terminating Equipment
3) PSE – Packet Switching Exchange
4) PAD – Packet Assembler/Disassembler



DTE – Data Terminal Equipment

End systems that communicate across the X.25 network.

Usually:

a) Terminals
b) Personal computers
c) Network hosts

Located on the premises of subscribers.

DCE – Data Circuit-Terminating Equipment

Communication devices that provide the interface between DTE and PSE.

Examples: Modems, packet switches

Located in the carrier's facilities.

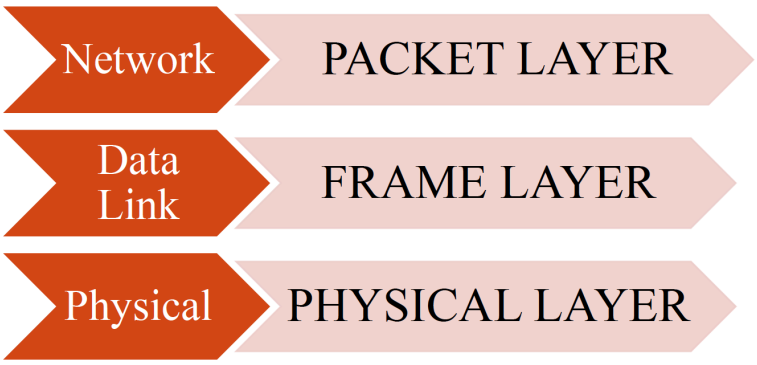
PSE (It is a part of DCE)

1) Core packet switche that form the main part of the X.25 network.
2) Transfer data from one DTE to another through the X.25 Packet-Switched Network (PSN).

PAD – Packet Assembler/Disassembler

1) Device located between a DTE and a DCE.
2) Performs three main functions:
   i. Buffering (temporary data storage)
   ii. Packet assembly
   iii. Packet disassembly

- X.25 protocol specifies 3 layers:



- Communication in this layer involves 3 phases:
  - Link Set Up
  - Data and Control transfer
  - Link disconnect
- These phases use different frames such as:
  - U-Frames
  - I-Frames
  - S-Frames

U-frame is used to set up and disconnect the connection between DTE and DCE
I-frames ---data transfer
S-frame is used for error control and flow control

Virtual Circuits in X.25

Two types of virtual circuits exist in X.25:

1) **Switched Virtual Circuits (SVCs)**
2) **Permanent Virtual Circuits (PVCs)**

SVC & PVC

Switched Virtual Circuits (SVCs)

Temporary connections used for irregular data transfers
They require that two DTE devices establish, maintain, and terminate a session each time the devices need to communicate

Permanent Virtual Circuits (PVCs)

For frequent and consistent data transfers
Do not require sessions established & terminated
DTEs begin transferring data whenever necessary because the session is always active

Usage in X.25

In X.25, SVC and PVC both are used.

The following 5 events occur:

1. link is set up between local DTE & DCE node and also between the remote DTE and DCE
2. VC is established between local and remote DTE
3. Data is transferred between two DTEs
4. VC is released
5. Link is disconnected

PLP Packet

A packet used to send data or control info over X.25 networks.

GFI (General Format Identifier) is a 4-bit field
(tell that whether it is data or control information)

LCN – Logical Channel Number(tells which channel the packet belongs to)

PTI (Packet Type Identifier)

defines the type of packet, viz. data packet or control packet
(there have 17 different types)

Virtual Channel ID Numbers

up to 4096 (2^{12}) multiplexed channels between each DTE and DCE
the calling and called hosts use different numbers

LCNs in X.25
Logical Channel Number identifies the VC at different sections of the network
A pair of LCN is defined when a VC is established between two DTEs

NIC (Network Interface Card)

Characteristics of NIC

- Physical network address (Hardware Component)
- Media Access Control (MAC) address
- MAC addresses are six bytes long
  - Example: 00-00-86-47-F6-65
- Network connection port
- Status indicator lights

Example: Wired NIC (Ethernet card), Wireless NIC (Wi-Fi adapter)

Mobile Phone System

Cordless Phones **vs Mobile Phones**

1. Mobile phones should not be confused with cordless phones.
2. Small political and marketing decisions shaped early mobile systems.

US vs Europe Differences

U.S.:

1) Mobile phones mixed with fixed-line numbers
2) Example: (212) 234-5678

Europe:

1. Each country created its own mobile system → incompatibility (fiasco)
2. Mobile phones use special area codes (like 800/900 numbers)
3. Prepaid mobile usage very high (up to 75% in some regions)

Generations of Mobile Phones

- ✔ 1G – Analog voice
- ✔ 2G – Digital voice
- ✔ 3G – Digital voice + data (Internet, email, etc.)

Early Mobile Telephone History(1G- Analog Voice)

1946: First car-based telephone system

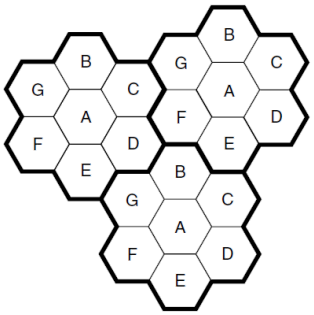
1. One large transmitter on a tall building
2. Single channel for both sending and receiving
3. Push-to-talk

1960s: IMTS (Improved Mobile Telephone System)

1) Supported 23 channels
2) Frequencies: 150 MHz – 450 MHz

Advanced Mobile Phone System (AMPS)

1. Developed by Bell Labs
2. Installed: 1982
   a) Variants:
   b) Europe → TACS
   c) Japan → MCS-L1
3. Retired: 2008



Cellular Concept

Geographic Region divided into cells
AMPS cells: 10–20 km
Digital systems → use more smaller cells

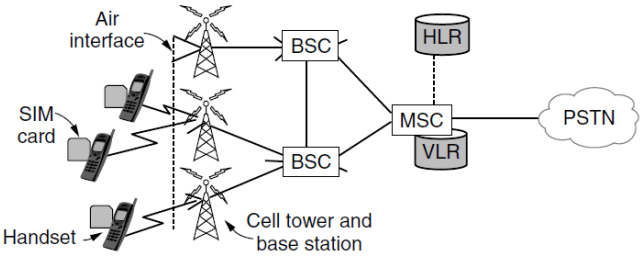
Key idea:

1) Frequency reuse → small cells + reuse in nonadjacent cells
2) Cells modeled as hexagons, grouped into clusters of 7
3) Frequency groups spaced with buffer cells to reduce interference

Cell Components

1) Base Station

1. Located at the center of each cell
2. Includes computer + transmitter/receiver + antenna



2) MSC / MTSO

1. Mobile Switching Center / Mobile Telephone Switching Office
2. Connects base stations
3. Acts like a telephone end office
4. Large systems may use multiple MSCs hierarchically

3) Handoff

When a mobile moves to another cell:

1) Signal weakens for the old cell
2) Base station transfers the call to the cell with strongest signal
3) Mobile is informed of new controller
4) Handoff time: ~300 ms

4) Channel Assignment

1. Done by the MSC
2. Base stations act mainly as radio relays

AMPS Channel Structure

- ✔ Uses FDM (Frequency Division Multiplexing)
- ✔ 832 full-duplex channels

Frequency allocations:

- ✔ 824–849 MHz → mobile → base
- ✔ 869–894 MHz → base → mobile
- ✔ Channel width: 30 kHz

Channel Types

1. Control channels (base → mobile): System management
2. Paging channels (base → mobile): Notify user of incoming call
3. Access channels (bidirectional): Call setup and assignment
4. Data channels (bidirectional): Carry voice, fax, or data

Second-Generation (2G) Mobile Phones: Digital Voice

- ✔ 1G: Analog voice
- ✔ 2G: Digital voice

Key Features of 2G

1) It provides capacity gains by allowing voice signals to be **digitized and compressed**
2) It improves security by allowing voice and control signals to be **encrypted**
3) Finally, it enables new services such as **text messaging**

Standardization

no worldwide standardization during the second

2G Technologies

- **DAMPS (Digital Advanced Mobile Phone System)**
  - ✔ coexists with AMPS
  - ✔ uses TDM to place multiple calls on the same frequency channel
- **GSM (Global System for Mobile communications)** – Europe
- **CDMA (Code Division Multiple Access)** – US

SIM Card

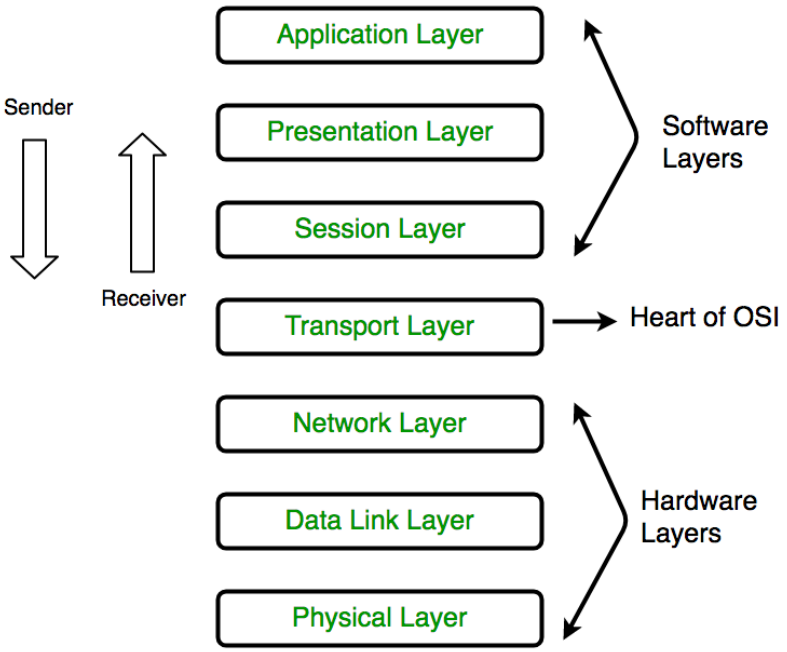
SIM card = Subscriber Identity Module

Network Components

1. **BSC (Base Station Controller)**
2. **BTS (Base Transceiver Station)**
3. **VLR (Visitor Location Register)**
   a) database of nearby mobiles that are associated with the cells it manages
4. **HLR (Home Location Register)**
   a) a database in the mobile network that gives the last known location of each mobile

Wk3-1-MAC Sublayer-Aloha...

Data Link Layer:



The **software layer** refers to the layers of a computer network that are implemented mainly using **software programs**. These layers define *how data is formatted, processed, and interpreted.*

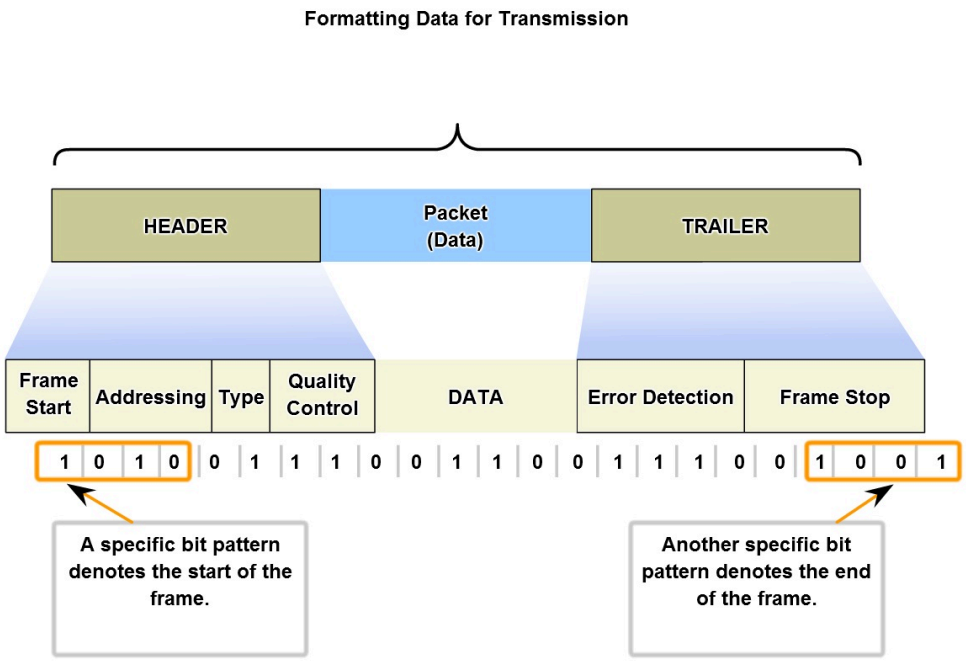
The **hardware layer** consists of the **physical components** used to transmit data.

The **Medium Access Control (MAC) sublayer** is a **part of the Data Link Layer** in the OSI model.

Data link layer

The main work of the **data link layer** is to make a physical transmission line appear **error-free** to the network layer.

1. It does this by dividing the incoming data into small units called frames and sending these frames one by one.
2. After receiving each frame, the receiver checks it for errors and sends an **acknowledgement (ACK)** if the frame is received correctly. If an error is found, the frame is sent again.



Addressing: Identifies the source and destination device using MAC addresses
Example: Sender MAC → Receiver MAC

Type: Indicates which upper-layer protocol the data belongs to
Example: IPv4, IPv6, ARP

Quality Control: Maintains reliable data transfer using acknowledgements and re-transmission
Example: Resending a frame if ACK is not received

Error Detection: Finds errors in transmitted data using CRC or checksum
Example: CRC field in an Ethernet frame

The **MAC (Medium Access Control) sublayer** controls **who can send data next** on a shared network channel.
It is especially important in **LANs**, particularly **wireless networks**, where multiple devices share the same medium and need rules to avoid collisions.

Example:
Ethernet (wired LAN): Uses **CSMA/CD** to **detect** and handle **collisions**.
Wi-Fi (wireless LAN): Uses **CSMA/CA** to **avoid collisions** before sending.

Types of Network Topology

1. Point to Point Topology
2. Mesh Topology
3. Star Topology
4. Bus Topology
5. Ring Topology
6. Tree Topology
7. Hybrid Topology

The Channel Allocation Problem
Multiple devices share a network channel. The **channel allocation problem** is about assigning channels efficiently to reduce interference and maximize usage.

Two Schemes of Channel Allocation

1. Static Channel Allocation (SCA)
Channels are **permanently assigned** to users or cells.
Uses **FDM (Frequency Division Multiplexing)**.

Drawbacks:

1. Users may be inactive, wasting bandwidth.
2. Poor fit for computer networks where traffic fluctuates.
3. Peak-to-mean traffic ratio can be as high as 1000:1 → most channels idle most of the time.

Example: Fixed frequency assignment in cellular networks.

2. Dynamic Channel Allocation (DCA)
Channels are **assigned on demand**, depending on traffic.
More efficient than static allocation.
Assumes a model with N stations and several rules:

Key Condition for DCA:

1) Independent Traffic:
Each station generates frames independently at a rate A.
Once a frame is generated, the station waits until it is successfully transmitted.

2) Single Channel:
One channel is shared by all stations for sending and receiving.

3) Observable Collisions:
If two stations transmit simultaneously, the signals collide.
All stations detect the collision.
Collided frames must be retransmitted.
No other errors occur.

4) Contnuous or Slotted Time:
Continuous: Transmission can start anytime.
Slotted: Time is divided into slots; transmissions start at slot beginnings.
Slot can be idle (0 frames), successful (1 frame), or collision (>1 frame).

5) Carrier Sense or No Carrier Sense:
With carrier sense: Stations check if the channel is busy before transmitting.
Without carrier sense: Stations transmit without checking; success is determined later.

Multiple Access Protocols

These protocols determine **how multiple users share a common communication channel** without too many collisions. They are commonly used in **Dynamic Channel Allocation (DCA)** systems.

Popular DCA Protocol: ALOHA

Origin:
Developed at the **University of Hawaii** in the early 1970s.
Stands for **"Additive Links Online Hawaii Access"** (sometimes jokingly called "Advocates of Linux Open-source Hawaii Association").
Purpose: Allow multiple users to share a single channel where conflicts (collisions) may occur.

1. Pure ALOHA

Users transmit **whenever they have data**.
Collisions may happen if two frames overlap in time.
If a collision occurs: the frame is retransmitted after a random time.
Simple but inefficient: maximum channel utilization ~18%.

2. Slotted ALOHA

Time is divided into **discrete slots**, each equal to one frame duration.
Users **must wait for the start of a slot** to transmit.
Reduces collisions because frames are aligned to slots.

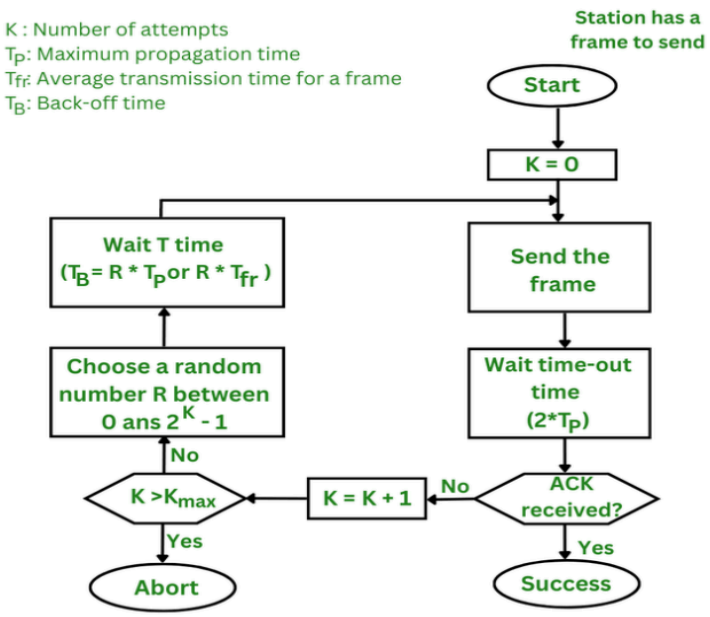
Requires synchronization:

One method: a special station emits a **timing signal** at the start of each slot.
Efficiency improves to ~37% maximum channel utilization.

Key Concept: Contention System
Any system where **multiple users compete for the same channel**.
Collisions are possible, so retransmission mechanisms are necessary.

Assumptions of Slotted ALOHA

1) **All frames are the same size.**
2) **Time is divided into equal slots**, each equal to the time to transmit one frame.
3) **Nodes transmit only at the start of a slot.**
4) **Nodes are synchronized** to the slot boundaries.



Carrier Sense Multiple Access (CSMA)

Definition: Stations **listen to the channel** before transmitting to reduce collisions.
Types of CSMA:

1. 1-Persistent CSMA
When a station has data:
Listens to the channel.
If idle → transmits immediately.
If busy → waits until idle, then transmits.
If a collision occurs → keep listening and transmit as soon as it becomes free.
Called **1-persistent** because it transmits **with probability 1** when the channel is free.

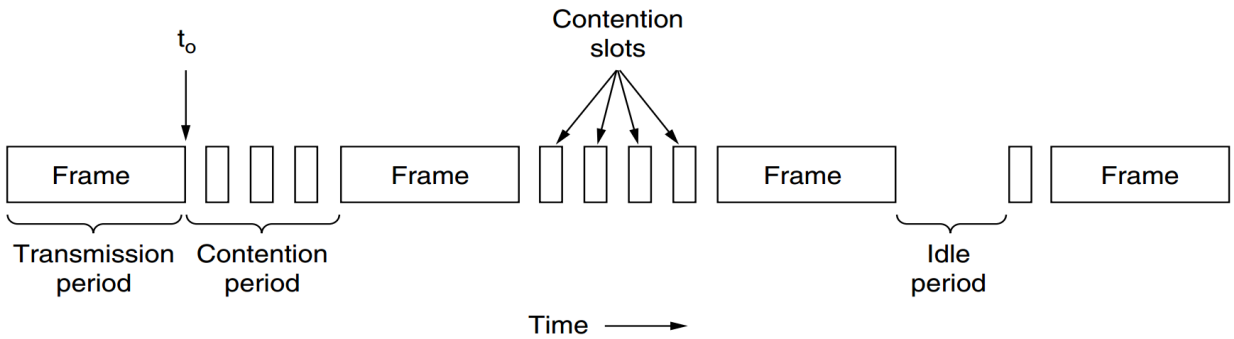
2. Non-Persistent CSMA
Station senses the channel:
If idle → transmits.
If busy → **waits a random time** before trying again (does not continuously sense).
Reduces collisions but may increase delay.

3. P-Persistent CSMA (for slotted channels)
When a station is ready:
Senses the channel.
If idle → transmits with probability **p**, or defers with probability **q = 1 − p**.
Repeats this process until the frame is transmitted or another station starts transmitting.
If a collision occurs → waits a random time and starts over.
If channel is busy → waits for the next slot and repeats the algorithm.

Collision detection: If two or more nodes transmit in the same slot, all nodes detect the collision before the slot ends.

CSMA with Collision Detection (CSMA/CD)

CSMA/CD is a multiple access protocol used in **wired LANs (Ethernet)** to reduce and handle collisions.



The **Bit-Map Protocol** is a **collision-free multiple access protocol** used in shared-channel networks. **Bit Map Protocol:**
A channel access method where each device has a bit (0 or 1) in a shared bitmap; if it wants to transmit, it sets its bit to 1, and then devices send data in order based on the bitmap, avoiding collisions.

Token Passing (Collision-Free Protocol)
Token Passing is a **collision-free multiple access protocol** used in shared networks. A network access method where a special data packet called a **token** controls which device can send data. Only the device that has the **token** is allowed to transmit. After sending, it passes the token to the next device.

Slide : Wk3-2-Ethernet-Hub-Switch

EEE 802 Standards Overview
Many network designs for **PAN, LAN, and MAN** have been standardized under the **IEEE 802** family.
Some standards became obsolete, but a few are still widely used today.

Major surviving IEEE 802 standards:

IEEE 802.3 → Ethernet (wired LAN)
IEEE 802.11 → Wireless LAN (Wi-Fi)
IEEE 802.15 → Bluetooth (wireless PAN)
IEEE 802.16 → Wireless MAN (WiMAX)

IEEE 802.11 (Wireless LAN)

First approved in **1997**
Operated at **1–2 Mbps** in the **2.4 GHz band**
The term **802.11** often refers to the **802.11x family**

IEEE 802.11b

Released: **September 1999**
Frequency: **2.4 GHz ISM band**
Maximum speed: **11 Mbps**

Problems:
Interference from cordless phones, microwave ovens, and other 2.4 GHz devices
Unregulated frequency band

IEEE 802.11a

Released around the same time as 802.11b
Frequency: **5 GHz**
Maximum speed: **55 Mbps**

Advantages: Less interference than 2.4 GHz
Disadvantages: Higher cost, Not compatible with 802.11b
Result: **Not very popular**

IEEE 802.11g

Released: **June 2003**
Frequency: **2.4 GHz**
Maximum speed: **55 Mbps**
Features:
Backward compatible with **802.11b**
Same interference issues as 802.11b
Better security (WPA, WPA2) -> Wifi Protected Access

IEEE 802.11n
Approved: **October 2009**
Frequency: **2.4 GHz and 5 GHz**
Speed: **54 Mbps to 600 Mbps**

Features:
Backward compatible with 802.11a/b/g
Supports **MIMO (Multiple Input Multiple Output)**
Higher speed and better range

IEEE 802.11ac
Operates only in the **5 GHz band**
Speed: **433 Mbps to several Gbps**

Features:
Up to **8 MIMO antennas**
Uses **beamforming** to focus signals toward devices
Very high performance

MIMO: Uses multiple antennas to send/receive data **simultaneously** → **higher speed**
Beamforming: Directs signal toward a device → **stronger, focused connection**

Ethernet
Ethernet is a widely used **LAN technology** standardized as **IEEE 802.3**.

Types of Ethernet
There are **two main kinds of Ethernet**:
1. **Classic Ethernet**
2. **Switched Ethernet**

Classic Ethernet
Speed: **3–10 Mbps**
Uses a **single long shared cable**
All computers are attached to the same cable
Uses **CSMA/CD**
Entire network is **one collision domain**

Physical structure:
Classic Ethernet cable **snaked around the building**, and all computers tapped into it.

Types of Classic Ethernet

1. Thick Ethernet (10Base5)
Cable looks like a **thick yellow hose**
Distance between taps: **2.5 meters**
Maximum cable length: **500 meters per segment**
Maximum devices: **100 machines per segment**

2. Thin Ethernet (10Base2)
Thinner and more flexible cable
Uses **BNC connectors**
Maximum cable length: **185 meters per segment**
Maximum devices: **30 machines per segment**

Cable Length and Repeaters
Each Ethernet segment has a **maximum unamplified length**
To build larger networks, **repeaters** are used

Ethernet rules:
No two transceivers can be more than **2.5 km apart**
Signal path must not pass through **more than 4 repeaters**

Switched Ethernet

Much faster and scalable
Uses **switches instead of shared cables**
Each device gets a **dedicated link**

Speeds:
Fast Ethernet → 100 Mbps
Gigabit Ethernet → 1000 Mbps
10 Gigabit Ethernet → 10,000 Mbps

Collision Domain

A **collision domain** is a network area where data packets can collide

In **Classic Ethernet**:
Entire network = **one collision domain**
In **Switched Ethernet**:
Each port = **separate collision domain**

Hub vs Switch, Repeater, Transceiver, and Modem

Repeater

A **physical layer device** that **receives, amplifies, and retransmits signals** in both directions.

Types of Repeater

1. Analog Repeater
2. Digital Repeater

Hub

A **physical layer device** that **broadcasts data to all connected ports**.

Types of Hub

1) **Active Hub** → Requires external power
2) **Passive Hub**
3) **Intelligent Hub**

Characteristics

- Broadcasts data to all devices
- Single collision domain
- Slow and outdated

Switch

A **data link layer device** that **forwards data only to the correct destination port**.

Characteristics

1. Uses MAC address
2. Each port = separate collision domain
3. Faster and efficient

Working Procedure of a Switch

1. Receives a frame
2. Reads destination MAC address
3. Checks MAC address table
4. Forwards frame to correct port
5. If unknown → floods the frame

Transceiver

A **TRANSmitter + reCEIVER** (mechanical device)

Types of Transceivers

1. RF Transceivers
2. Fiber Optic Transceiver
3. Ethernet Transceivers
4. Wireless Transceivers

Key Features / Functions
Modulation & Demodulation
Signal Amplification
Frequency Conversion
Signal Encoding/Decoding

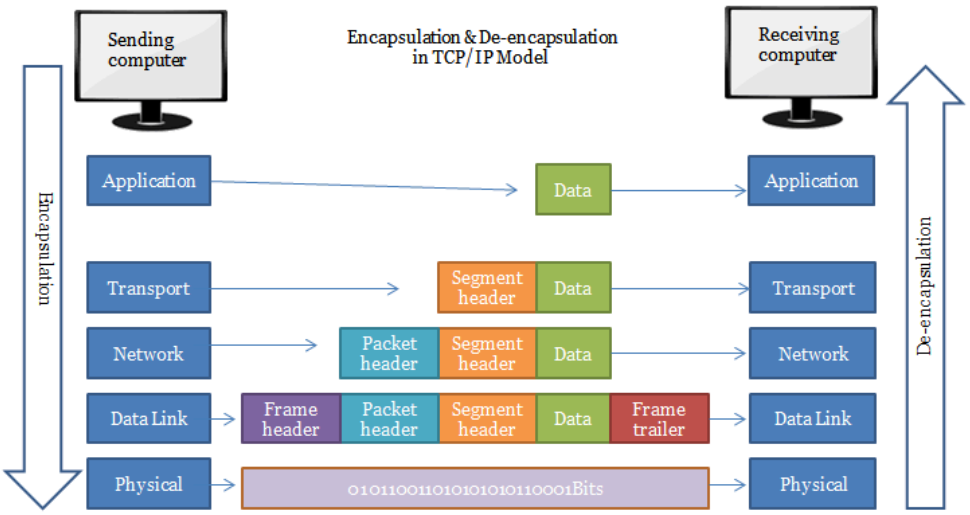
Modem
Modem = Modulation + Demodulation

Key Concept Question
Why is repeater called Level-1 relay?
→ Because it works at the **Physical Layer (Layer 1)** and only deals with signals, not data.

Hub vs Switch vs Router

Feature	Hub	Switch	Router
Layer	Physical (Layer 1)	Data Link (Layer 2)	Network (Layer 3)
Function	Broadcasts to all ports	Sends to specific port	Routes between networks
Address Used	None	MAC address	IP address
Collision Domain	Single	Separate per port	Separate networks
Efficiency	Low	High	Very high
Security	None	Moderate	High
Usage	Old/simple networks	LAN networks	Internet & inter-networking

Slide : Wk4-1-Data Link Layer-Stop and wait- Sliding window



DATA LINK LAYER DESIGN ISSUES

The **Data Link Layer** (Layer 2 of the OSI model) is responsible for **reliable communication** between two directly connected nodes.

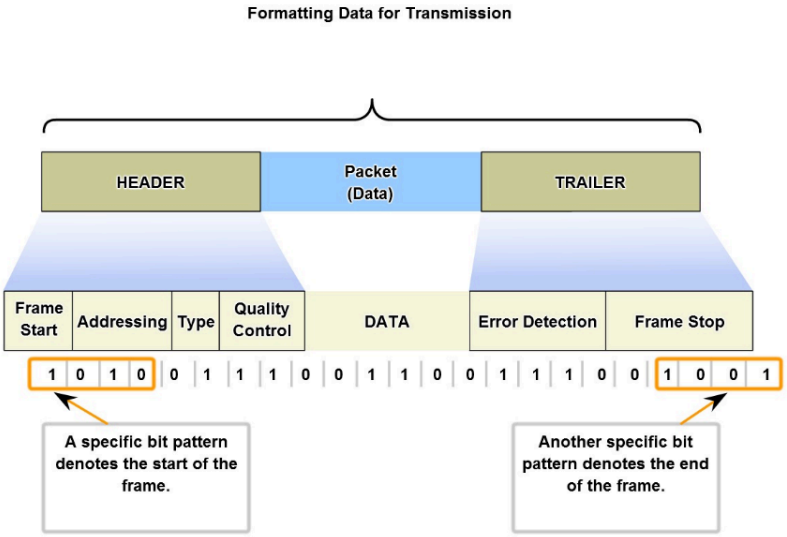
The main task of the **Data Link Layer** (Layer 2 of the OSI model) is to make a noisy physical transmission medium appear reliable to the Network Layer(Layer 3 of the OSI model).
The physical layer(Layer 1) only transmits raw bits, and errors may occur due to noise, interference, or signal distortion.

To handle this, the Data Link Layer:

1. Divides incoming big data into small units called **frames**.
2. Adds control information (such as sequence numbers and error detection codes like CRC).
3. Sends frames one after one. They are also numbered so the receiver knows the correct order.
4. Waits for an **acknowledgement (ACK)** from the receiver.
5. Retransmits a frame if it is lost or corrupted.

The receiver checks each frame for errors and sends an ACK (acknowledgement) if the frame is correct. If corrupted sends NAK. If sender does not receive ACK within a time limit:, the sender retransmits the frame.
This process is called **Automatic Repeat reQuest (ARQ)**.

By using framing, error detection, acknowledgements, and retransmission, the Data Link Layer ensures reliable communication between two directly connected nodes and makes the transmission channel appear free of undetected errors.



Main Design Issues

1. **Providing a well-defined service interface to the Network Layer**

Hides transmission details
Provides reliable data transfer service

2. **Dealing with transmission errors**

Detects and handles errors during transmission

Regulating the flow of data so that slow receiver are not swamped by faster senders

Prevents a fast sender from overwhelming a slow receiver is Known as **flow control**
If a fast sender sends data to a slow receiver, the receiver's buffer may overflow.

Flow control ensures:

1) Sender does not send too fast
2) Receiver can process data safely

Common methods:

1) Stop-and-Wait
2) Sliding Window Protocol

Packets vs Frame
Packet: Data unit at the **Network Layer**
Frame: Data unit at the **Data Link Layer**

Sending Machine Receiving Machine
Packet ----> Frame ----> Packet

Frame Structure
| Header | Payload (Data) | Trailer |

Payload: Actual data being transmitted

Framing

Framing is the process of dividing a continuous bit stream into **frames**.

Common Framing Methods

1. Byte Count
Header contains the **number of bytes** in the frame, uses length field.
Simple but errors in count can cause loss of synchronization and full data corrupted.

2. Flag Byte with Byte Stuffing
Frames start and end with a **flag byte**
Two consecutive flags indicate **end of one frame and start of next**

Problem: Flag byte may appear in data
Solution: Insert an **ESC (escape) byte** before accidental flag bytes
This technique is called **byte stuffing**

3. Flag Bits with Bit Stuffing
Frames start and end with a **special bit pattern**
Flag pattern: **01111110 (0x7E)**
If five consecutive 1s appear in data, a **0 is inserted**
Receiver removes the extra 0

4. Physical Layer Coding Violation

1. This method uses special signal patterns that are not valid data signals.
2. Certain signal levels or transitions are reserved to indicate the start and end of a frame.
3. Since these patterns never appear in normal data, they clearly mark frame boundaries.

Advantage:

1. No need for stuffing.
2. Clear and reliable framing.

Limitation: Depends on the type of physical layer encoding scheme being used.

Types of Errors

Errors during transmission can be:

1. **Single-bit errors** – only one bit is changed
2. **Multiple-bit errors** – more than one bit is changed
3. **Burst errors** – a sequence of bits is corrupted

1. Error-Detecting Codes

Detects if data has been corrupted.

Common methods:

1. Parity bits – adds 1 extra bit to make total 1s even or odd.
2. Checksum – sum of all data units, sent with data.
3. Cyclic Redundancy Check (CRC) – more reliable, uses polynomial division.

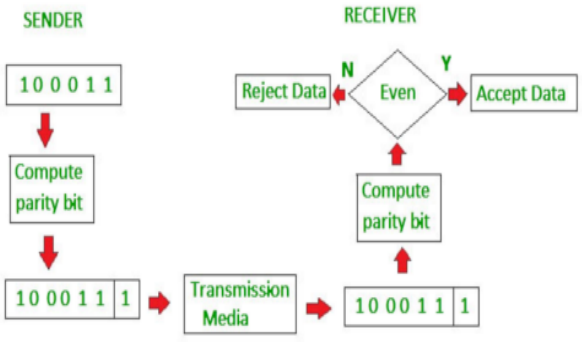
Limitation: Only detects errors; does not correct them.

2. Error-Correcting Codes

Detects and corrects errors without retransmission.
Adds redundant bits so the receiver can identify and fix errors.

Popular error-correcting codes:

1. Hamming codes.
2. Binary convolutional codes.
3. Reed-Solomon codes.
4. Low-Density Parity Check codes.



Error Detection Techniques

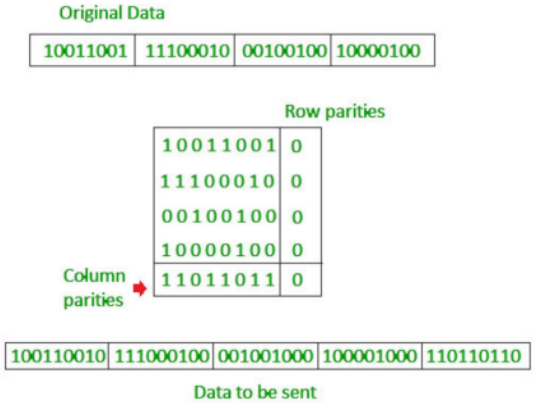
Common Methods

- ✔ Parity Check
- ✔ Checksum
- ✔ Cyclic Redundancy Check (CRC)

Parity Check

1. Simple Parity Check

a) Adds one parity bit (even or odd)
b) Detects **single-bit errors**
c) Cannot detect all multiple-bit errors



2. Two-Dimensional Parity Check

a) Data arranged in rows and columns
b) Parity bits added for both
c) Better detection than simple parity

Checksum (Error-Detecting)

1. Data is divided into equal-sized words.
2. Words are added together, and the complement is sent as the **checksum**.
3. Receiver adds all words (including checksum); if result is correct → no error; otherwise → error detected.

Limitation: Can miss some multiple-bit errors.

Cyclic Redundancy Check (CRC) (Error-Detecting)

1) Data is treated as a binary polynomial.
2) Divided by a **generator polynomial**; remainder (CRC) is appended to data.
3) Receiver divides received data; if remainder = 0 → no error; otherwise → error.

Advantage: Detects single, double, and burst errors reliably.

Flow Control and Error Control

Flow control deals with problem that sender transmits frames faster than receiver can accept, and solution is to limit sender into sending no faster than receiver can handle

Simple Stop-and-Wait Protocol

Purpose: Ensures reliable data transfer between sender and receiver.

Sender Rules

1. Send **one data packet at a time**.
2. Wait for **acknowledgement (ACK)** from the receiver before sending the next packet.

Receiver Rules

1. Receive the data packet and **deliver it to the upper layer**.
2. Send an **acknowledgement (ACK)** after consuming the packet to control flow.

Problems in Simple Stop-and-Wait Protocol

1 – Infinite Wait

1. **Sender:** Waits for ACK.
2. **Receiver:** Waits for data.
3. **Problem:** If a packet or ACK is lost, both **wait indefinitely**, leading to a deadlock.

2 – Lost ACK or Data

1. **Sender:** Waits for ACK forever if it never arrives.
2. **Receiver:** May have already received or missed the data.
3. **Problem:** Sender may **never know** the status of the packet.

3 – Delayed ACK Problem
Scenario: ACK arrives **after a long delay**.
Problem: Sender may mistake a **late ACK** as acknowledgment of a **newer packet**, causing **duplication or misdelivery**.

Stop-and-Wait ARQ (Automatic Repeat Request)

Purpose:

1. Ensures reliable data transfer.
2. Handles lost data, lost ACKs, and delayed ACKs.
3. Combines error control and flow control.

Key Features

1. Sequence Numbers
   - Each data frame is numbered (0 or 1 for Stop-and-Wait).
   - Each ACK contains the sequence number of the next expected frame.
   - Helps detect duplicates.

2. Timeout and Retransmission
   - If the sender does not receive an ACK within a certain time, it retransmits the frame.

3. Receiver Behavior
   - Accepts a frame only once.
   - Sends ACK with the sequence number of the next expected frame.
   - Discards duplicates.

How Problems are Solved

Problem	Solution in Stop-and-Wait ARQ
Lost Data	Retransmission occurs after timeout.
Lost ACK	Sender retransmits, receiver uses sequence numbers to avoid duplicate delivery.
Delayed ACK	Sequence numbers prevent confusion between old and new ACKs.

Example Flow

1. Sender A → Receiver B: Frame 0
2. Receiver B: Accepts Frame 0, sends ACK 1 (next expected frame = 1)
3. Sender A: Receives ACK 1, sends Frame 1
4. If Frame 0 or ACK 0 is lost, sender retransmits Frame 0, and receiver discards duplicates using sequence numbers.

Sliding Window ARQ (Automatic Repeat Request)

Sliding Window ARQ is an **advanced error control and flow control mechanism** used at the **data link layer**. It improves efficiency compared to **Stop-and-Wait ARQ**.

Problems Solved Using ARQ

1. Error control
2. Flow control
3. Synchronization

Main Idea

Instead of sending **one frame at a time** (Stop-and-Wait), the sender can **send multiple frames** before needing an acknowledgement.
The number of frames that can be sent without ACK is called the **window size**.
Both sender and receiver maintain a **window** of frames they can send or receive.

Sliding Window ARQ

Frames and acknowledgements are numbered using **sequence numbers**.

Sender:

1. Maintains a list of sequence numbers it is allowed to transmit, called the **sending window**.
2. A sending window of size **N** means the sender can send up to **N frames** without waiting for an ACK.
3. A window size of **N** also implies buffer space for **N frames**.

Receiver:

1. Maintains a list of sequence numbers it is prepared to receive, called the **receiving window**.
2. Sends ACKs indicating the **next expected frame**.

Sequence Numbers:

For an **n-bit sequence number**, the numbers range from 0 to 2n−12^n - 12n−1.
Maximum window size is 2n−12^n - 12n−1 (not 2n2^n2n).

Example:
Though sender transmit 7 data but ACK 3 means the receiver has correctly received frames 0, 1, and 2, and is now ready to receive frame 3 (and the rest of the frames within the window).

Sliding Window ARQ Variants

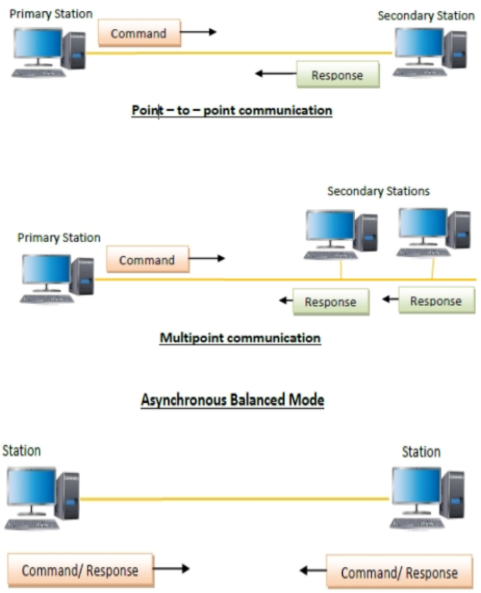
Go-Back-N ARQ: On an error in Nth frame, go back and retransmit N frames starting from the frame with the error.
Selective Reject ARQ: On an error in any frame, retransmit only the rejected frame, and the receiver arranges frames in order.

Slide : Wk4-2-HDLC

HDLC (High-Level Data Link Control)
HDLC is a **data link layer protocol** used for **reliable communication** between network nodes.
Provides **framing, error detection, flow control, and link management**.

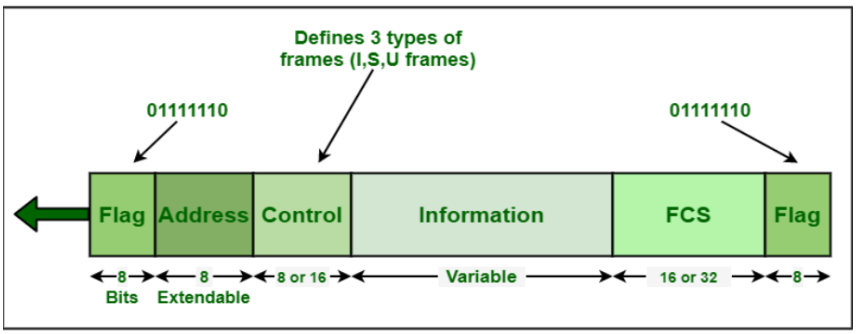
HDLC Transfer Modes

1. Normal Response Mode (NRM)



Two types of stations:
Primary station: Sends commands
Secondary station: Responds to commands
Usage: Point-to-point and multipoint networks
Only the **primary station can initiate transmission**

2. Asynchronous Balanced Mode (ABM)
Balanced configuration: All stations are equal
Each station can **send and respond to commands**
Usage: Only point-to-point networks
Supports **full-duplex communication**



Basic Frame Structure

HDLC Basic Frame Structure

Flag Field

8-bit pattern: **01111110 (0x7E)**

Marks **start and end of a frame**

Serves as a **synchronization pattern**

Cannot appear elsewhere in the frame

	1	2	3	4	5	6	7	8
I: Information	0	N (S)			P/F	N (R)		
S: Supervisory	1	0	S		P/F	N (R)		
U: Unnumbered	1	1	M		P/F	M		

N (S): Send Sequence Number
N (R): Receive Sequence Number
S: Supervisory Function Bits
M: Unnumbered Function Bits
P/F: Poll/Final Bit

Control Field Format

Address Field

1 or more bytes

Identifies the **receiver address**

Control Field

Indicates **frame type** and controls flow/error management

Payload Field

Carries **user data** from the network layer that vary in different network.

FCS Filed

It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

Types of HDLC Frames

1. **I-frame (Information frame):**
   i. Carries user data from the network layer.
   ii. Includes flow and error control information (piggybacked).
   iii. Control field first bit = 0.

2. **S-frame (Supervisory frame):**
   i. Contains no user data.
   ii. Used for flow and error control only.
   iii. Control field first two bits = 10.

3. **U-frame (Unnumbered frame):**
   i. Used for link management and miscellaneous functions.
   ii. May contain user data if needed.
   iii. Control field first two bits = 11.

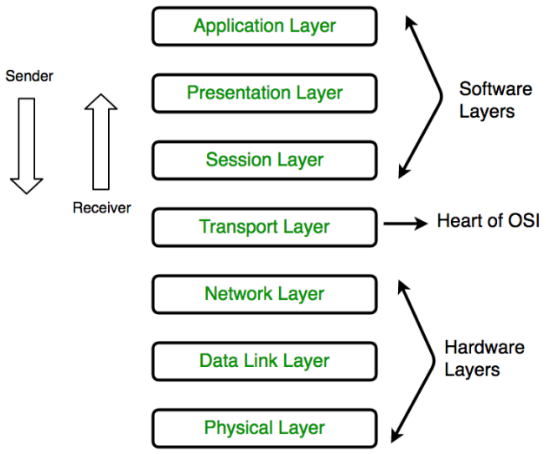
Notes:
I-frame: Piggybacks control info with user data
S-frame: Only controls transmission (ACK, flow control)
U-frame: Handles miscellaneous tasks like link setup, termination, or mode changes

Slide: Wk5-1-Layer Role-Classful

Network Layer (Layer 3 – OSI Model)
The network layer is responsible for moving data packets across multiple networks from a source host to a destination host.
The network layer makes sure your data finds its way from your computer to another computer, even if it has to cross many different networks in between.
Key protocol: Internet Protocol (IP)

Key Functions of the Network Layer

1. Assigning Logical Address
2. Fragmentation and Reassembly of packets
3. Packetizing
4. Logical Subnetting
5. Host-to-Host delivery
6. Network Address Translation
7. Forwarding
8. Routing



1. Assigning Logical Address

Logical Addressing refers to assigning an IP address (IPv4 or IPv6) to a device so it can communicate over a network.
Unlike a MAC address:

1. A **MAC address** is physical and usually fixed to the hardware.
2. An **IP address** is logical and can change depending on the network.

Why it is called "logical"
Because it is assigned based on network structure and rules, not burned into hardware.

Key Characteristics

1. **Hierarchical structure:** An IP address has:
   i. Network portion
   ii. Host portion

2. This hierarchy allows routers to determine where a device is located.

3. **Changeable**
   A device may have different IP addresses on different networks (for example, at home vs university).

Importance of Logical Addressing

1. **Communication across networks:** It allows devices on different networks to communicate.

2. **Routing:** Routers use the network portion of the IP address to forward packets efficiently.

2. Fragmentation and Reassembly

Networks have a **Maximum Transmission Unit (MTU)**
If a packet is too large:

1. Network layer **fragments it** into smaller packets
2. Adds headers for identification and sequencing
3. Destination **reassembles fragments** into the original packet

3. Packetizing

Packetizing is the process where the **network layer**:

1. **Encapsulates** data from the transport layer by adding an IP header (source and destination IP).
2. Sends the packet to the data link layer.
3. At the destination, it **decapsulates** the packet and delivers the payload to the upper layer.

4. Logical Subnetting

Divides a large network into **smaller subnets**
Benefits:

1. Improves performance (reduces congestion)
2. Enhances security (isolates parts of the network)

Uses **subnet masks** for address allocation

5. Host-to-Host Delivery

The network layer ensures data is delivered from the source host to the destination host across one or multiple networks.

6. Network Address Translation (NAT)

NAT allows multiple devices in a private network to share a single public IP address for internet access.

1. Translates **private IP** → **public IP** for outgoing traffic
2. Translates **public IP** → **private IP** for incoming traffic

Benefits:

1. Conserves IPv4 addresses
2. Hides internal IPs for security

7. Forwarding

Routers transfer packets toward their destination
Uses **routing tables** and **destination IP** to decide path
Supports **unicast** (single destination) and **multicast** (multiple destinations)

8. Routing

Determines the **best path** for data from source to destination
Uses **routing protocols** to coordinate routers
Ensures **efficient and reliable data delivery**

Duty of Routers:

1. Only read the IP header to forward packets.
2. Do not change source/destination IP addresses.
3. Do not decapsulate packets (except for fragmentation if needed).

IP Address

An IP address is a unique number assigned to a device to identify it on a network and enable communication; it may be dynamic or static and changes when moving between networks.

1. Uses the **Internet Protocol (IP)**.
2. Ensures data reaches the **correct destination**.

Types of IP Address

1. Based on Addressing Scheme (IPv4 vs. IPv6)
2. Based on Usage (Public vs. Private)
3. Based on Assignment Method (Static vs. Dynamic)

Public vs Private IP Address

Private IPv4 addresses

RFC 1918 name	IP address range	Host ID size
24-bit block	10.0.0.0 – 10.255.255.255	24 bits
20-bit block	172.16.0.0 – 172.31.255.255	20 bits
16-bit block	192.168.0.0 – 192.168.255.255	16 bits

IPv4 Address

1. **Length:** 32 bits
2. **Address space:** 2^{32} = 4,294,967,296 unique addresses
3. **Example (binary → decimal):**

Binary: 11000000 10101000 00000000 00000001
Decimal: 192.168.0.1

Public IP

1. Globally unique
2. Routable on the Internet

Private IP

1. Used within local networks (LAN)
2. Not routable on the Internet
3. Defined by **RFC 1918**

Address space
An **address space** is the total set or range of discrete (separate) addresses available in a system

The address space in a protocol That uses N-bits to define an Address is: $\mathbf{2^N}$

Note on RFCs
RFC (Request for Comments) are **technical documents** published by the **IETF** under the **Internet Society (ISOC)**.
They define **Internet standards**, including IP addressing rules.

How IP Addresses Are Assigned

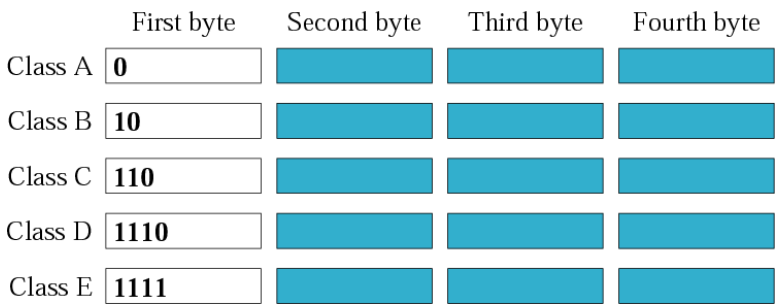
1. **ISP / Internet (Public IP):**
   a) Assigns a public IP to your router
   b) This public IP is what websites and external servers see
2. **Router / Local Network (Private IP):**
   a) Acts as a **DHCP server**
   b) Assigns a private IP to your laptop, phone, or tablet when they connect

A **DHCP server** is a server that automatically assigns IP addresses and other network settings to devices in a network.
DHCP stands for

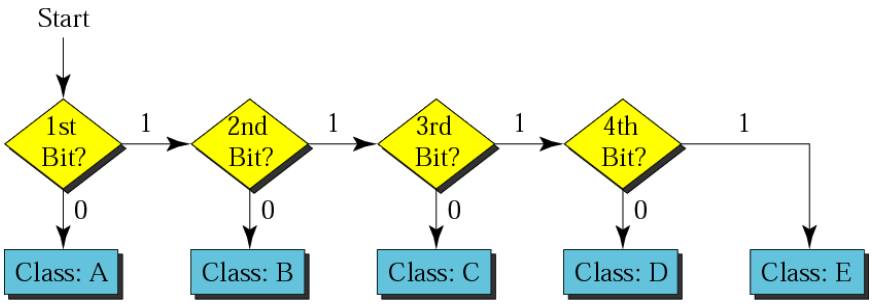
Dynamic Host Configuration Protocol — a protocol used with the Internet Protocol at the Network Layer of the OSI model.

Classful Addressing

Classful addressing is an IP addressing scheme where the IPv4 address space (32 bits) is divided into five classes (A, B, C, D, E) to allocate networks efficiently.



Finding the address class:



Finding the class in decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	**0 to 127**			
Class B	**128 to 191**			
Class C	**192 to 223**			
Class D	**224 to 239**			
Class E	**240 to 255**			

Port Number

A **port number** identifies a **specific service or application** on a device.
Works together with an IP address to direct data to the correct program.
Example: Email client, File server, Game

Format: IP_address : Port_number Example: **158.128.1.108** : **25**

1. 158.128.1.108 → IP address of the device
2. 25 → Port number (used by SMTP email service)

Port range: 0 – 65535 (16-bit integer)

Well-known ports (default): 1 – 1024

Example services:
Ping → 7, DHCP → 68, HTTP → 80, SMTP → 25

IP Address and Port Together

Think of it like a **street address and apartment number**:
IP → Street / building
Port → Apartment / service inside the building

Example:

192.168.0.5:80
192.168.0.5 → Device on the network
80 → Web server running on that device

NetID and HostID

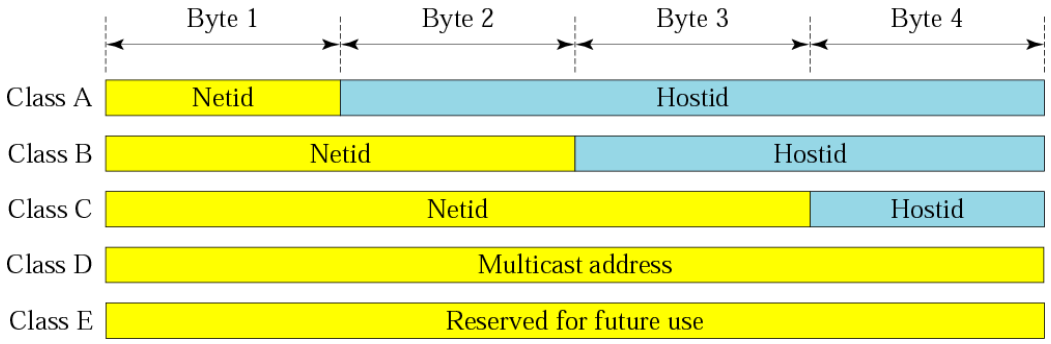
Every IP address (IPv4) has **two parts**:

1. **NetID** – Identifies the **network**
2. **HostID** – Identifies a **specific device (host)** on that network

Example (Class C IP: 192.168.1.10)

NetID: 192.168.1 → network part
HostID: 10 → host part

Key idea:
NetID + HostID = full IP address, which tells routers **which network** and **which device** to send data to.



Network ID (NetID): Identifies the network. All devices in the same network share the same NetID.

Host ID (HostID): Identifies a specific device on that network. Must be unique in the network.

Examples:
Class C (192.168.1.10):
NetID = 192.168.1 and HostID = 10
Key Idea:
NetID = "Which network?"
HostID = "Which device in that network?"

NetID and HostID in Routing

Routers use NetID:
To forward packets to the correct network.
Example: Packet to 192.168.2.10 → router checks NetID = 192.168.2 → forwards to that network.

Devices use HostID:
Once the packet reaches the network, devices use HostID to deliver the packet to the exact device.

Math:

How to Determine the Class

Look at the first octet of the IP address.

1–126 → Class A
128–191 → Class B
192–223 → Class C
224–239 → Class D (multicast)
240–255 → Class E (experimental)

Only Class A, B, C are used for normal host addressing.

How to Determine Default Subnet Mask

Each class has a fixed default mask (classful addressing):
Class A → 255.0.0.0 → /8
Class B → 255.255.0.0 → /16
Class C → 255.255.255.0 → /24

How to Determine Network Address

Steps:

1. Identify the class
2. Keep the network portion
3. Set all host bits to 0

Network portions:

1. Class A → first 1 octet
2. Class B → first 2 octets
3. Class C → first 3 octets

Example (Class B):

150.25.67.89
Network = 150.25.0.0

How to Determine Broadcast Address

Set all host bits to 1 (255).
Example (Class B):
Broadcast = 150.25.255.255

How to Determine Number of IP Addresses

Formula:
Total IPs = 2^(number of host bits)

Host bits:

1. Class A → 24 host bits
   Total = 2^24 = 16,777,216
2. Class B → 16 host bits
   Total = 2^16 = 65,536
3. Class C → 8 host bits
   Total = 2^8 = 256

Usable IP addresses:
Usable = Total − 2
(One for network address, one for broadcast)
Example (Class C):
256 − 2 = 254 usable IPs

Not In Slide(Here Given For Better Understanding)

NAT (Network Address Translation)
Your router replaces private IP with **public IP** when sending to the Internet.
Keeps a mapping table: internal device IP + port → external IP + port.
Server responds to public IP; router uses mapping to forward to correct internal device.

Key Points:
HostID of your device never go out from your network.
Routers on Internet only see public IP (NetID) of your router.
ISP can track thousands of users with efficient mapping tables (port + IP).

Routing Across the Internet

1. **Next Hop Routing:** Routers only know the next hop, not full path.
2. **Routing Tables:** Contain network prefixes, not every host IP.
a. Example: 203.0.113.0/24 covers 256 host IPs.
3. **TTL (Time to Live):** Prevents infinite loops.
4. **Routing Protocols:** BGP, OSPF, RIP ensure best path and prevent loops.

How a packet reaches Facebook:

1. Laptop → home router → ISP router
2. ISP router → next-hop router → ... → Facebook network
3. Local router at Facebook → exact server (HostID inside network)
4. Each router forwards based on **NetID**, not individual host.
5. HostID inside destination network used only locally.

Step-by-Step Packet Flow (Request → Server → Response)

1) **Device sends request** (private IP, HostID + NetID)
2) **Router (NAT)** replaces private IP with public IP; maps internal device to port
3) **Packet travels Internet** using **destination NetID**; HostID ignored
4) **Server receives packet**; responds to router's public IP
5) **Router receives response**; uses NAT table to forward to correct internal HostID
6) **Device receives response**; communication complete

Key Notes:
Public IP of router stays same across Internet; private IP (HostID) never leaves LAN.
Routers and ISPs do not track individual devices; NAT handles mapping.
HostID is only relevant inside your network.

Why HostID is Needed Inside Your Home Network
Your router sees many devices Suppose you have a laptop, phone, and smart TV on the same network:

Device	Private IP (HostID part)
Laptop	192.168.1.5 (HostID=5)
Phone	192.168.1.6 (HostID=6)
TV	192.168.1.7 (HostID=7)

All these devices share the same **NetID** = 192.168.1
Without HostID, the router **cannot distinguish which device** to send a response to.

NAT table alone is not enough without HostID
NAT maps **private IP + port → public IP + port**
If devices didn't have unique HostIDs (part of their private IP), **the router wouldn't know which internal device corresponds to which private IP**.
HostID is essentially the unique identifier for each device **inside your LAN**.

Why ISPs don't need HostID
The ISP sees **only your router's public IP**.
It doesn't need to know about individual devices in your LAN.
NAT handles the translation: public IP + port → internal private IP + port (HostID helps the router do this mapping).

Analogy: Imagine a building with multiple apartments (HostIDs) on the same street (NetID).
The post office (ISP/Internet) only sends the mail to the street (NetID).
The building manager (router) looks at the apartment number (HostID) to deliver mail to the correct apartment.

Key Point
HostID is useless outside your LAN → it has no role on the Internet.
HostID is absolutely necessary inside your LAN → your router cannot deliver packets to the correct device without it.

Slide: Wk5-2

Problems with Classful IP Addressing

Class A Address Wastage

1. Class A provides $2^{24} \approx 16$ million host addresses per network.
2. Most organizations do not need this many addresses.
3. As a result, millions of addresses are wasted in each Class A network.

Class B Address Wastage

1. Class B provides $2^{16} = 65{,}536$ host addresses per network.
2. Many organizations need far fewer than this.
3. Hence, a large number of Class B addresses are wasted.

Class C Address Limitation

1. Class C provides $2^{8} = 256$ addresses (254 usable hosts).
2. This is too small for most organizations.
3. Organizations needing slightly more than 254 hosts cannot use Class C efficiently.

Class D Address Usage

1. Class D addresses are not used for host addressing.
2. They are reserved only for multicasting.

Class E Address Wastage

1. Class E addresses are reserved for experimental and special purposes.
2. They are not used in normal networking.
3. Therefore, most of the Class E address space is wasted.

Overall Conclusion
Classful addressing causes significant IP address wastage.
It lacks flexibility in allocating addresses based on actual need.
This problem led to the introduction of **CIDR (Classless Inter-Domain Routing)**, which allows efficient and flexible IP allocation.

Assignment in Classful Addressing

In **classful addressing**, the **organization is assigned the network address**.
Meaning: the organization gets the **entire block of IPs** for its use.

Example:
Organization assigned 172.16.0.0/16
They can assign IPs to their devices: 172.16.0.1 → 172.16.255.254

Network address is the First Address of the Block

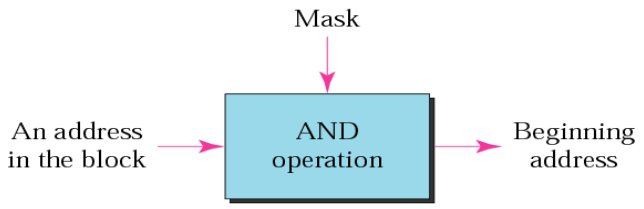
1. Every network (or subnet) has a range of IP addresses.
   Example: 192.168.1.0 – 192.168.1.255
2. The network address is the very first IP in the range.
   In the example above, 192.168.1.0 is the network address.
3. All other IPs in the range (192.168.1.1 – 192.168.1.254) are usable by devices (hosts).
4. The network address itself is not assigned to any device.
   It just identifies the network.

Block of addresses = 192.168.1.0 → 192.168.1.255
Network address = 192.168.1.0 ← first address in the block
Usable host addresses = 192.168.1.1 → 192.168.1.254
Broadcast address = 192.168.1.255 ← last address in the block
So "first address of the block" just means **the first IP of the network**, which represents **the entire network**.

What is a Mask?

A **mask** is a **32-bit binary number**. Its purpose is to **separate the network part (NetID) from the host part (HostID)** in an IP address.

When you apply the mask to an IP address, you get the **network address**.



How Mask Works
You use a **bitwise AND operation** between the IP address and the mask.

Formula:
Network Address = IP Address AND Mask
"AND" means:
1 AND 1 = 1
Anything else = 0
This keeps the **network bits** the same and sets all **host bits to 0**.

Network Address or Block Address
Network address = first address in the network block.
Can be found by applying the mask to any IP in the block.
It retains the NetID and sets HostID bits to 0.

Example:
IP = 192.168.1.130
Default Class C mask = 255.255.255.0 (/24)

Binary:
IP: 11000000.10101000.00000001.10000010
Mask: 11111111.11111111.11111111.00000000
AND 11000000.10101000.00000001.00000000

Network address = 192.168.1.0
The **first address of the block** = 192.168.1.0
The **network bits (NetID)** = 192.168.1
The **host bits (HostID)** = 0

Default Masks (Classful)
Class A: Mask 255.0.0.0 → 8 network bits, 24 host bits
Class B: Mask 255.255.0.0 → 16 network bits, 16 host bits
Class C: Mask 255.255.255.0 → 24 network bits, 8 host bits

Default mask applied = used to find the network address
For example:
IP 10.5.6.7 (Class A) → default mask 255.0.0.0 → network address = 10.0.0.0
IP 172.16.5.10 (Class B) → mask 255.255.0.0 → network = 172.16.0.0
IP 192.168.1.10 (Class C) → mask 255.255.255.0 → network = 192.168.1.0

What is Subnetting?

Subnetting means dividing a larger network into **smaller networks** (subnets). Done by **borrowing bits from the host part** and adding them to the network part.

Why we do subnetting?

1. To organize networks efficiently
2. To reduce wasted IP addresses
3. To control traffic within smaller subnetworks

How Subnet Address is Found

Like network address, we calculate **subnet address** by applying the **subnet mask** to an IP address using the **AND operation**.

Formula: Subnet Address = IP Address AND Subnet Mask

Straight Method (Binary)

1. Convert **IP address** and **subnet mask** into **binary**.
2. Perform **bitwise AND** (1 AND 1 = 1, else 0).
3. Result = **subnet address** (first address of that subnet).

Example:

IP: **200.45.34.56** and Subnet Mask: **255.255.240.0**

Step 1: Convert to binary

IP: 11001000.00101101.00100010.00111000
Mask: 11111111.11111111.11110000.00000000

Step 2: AND operation

Result: 11001000.00101101.00100000.00000000

Step 3: Convert back to decimal

Subnet Address = 200.45.32.0
This is the **first address of the subnet**.

Shortcut Method

1. Instead of converting the full address to binary:
2. Look at the **octet where subnetting occurs**
3. Identify **block size** = 256 − subnet mask in that octet
4. Find which **multiple of block size** the host falls into

IP: **200.45.34.56** and Subnet Mask: **255.255.240.0**
For the example above:

1. Subnet mask = 255.255.240.0 → third octet is 240
2. Block size = 256 − 240 = 16
3. IP third octet = 34 → subnet = 32 (nearest **multiple of 16 less than 34**)
4. Subnet address = 200.45.32.0

Important Rules
Number of subnets = power of 2
Borrow n bits → 2^n subnets
Subnet mask determines **which bits are network bits** now
Host bits = remaining bits → usable host addresses

Classless Addressing (CIDR)

CIDR (Classless Inter-Domain Routing) allows any number of network bits, not fixed by class
Not limited to A, B, C

Example:
IP: **200.45.34.56/20**
Here /20 → first 20 bits are **network/subnet**, last 12 bits are **host**
Works the same way as subnetting

What actually happens in subnetting
In subnetting, we **borrow bits from the host portion** (HostID) and add them to the network portion.

We do NOT reduce the original NetID. Instead, we **extend the network portion by taking some host bits**.

How modern addressing solves this (CIDR / subnetting)

CIDR (Classless Inter-Domain Routing) allows **any number of bits for the network ID**, not fixed 8, 16, or 24.

Examples:
/16 → network 16 bits, host 16 bits → large organizations
/20 → network 20 bits, host 12 bits → smaller organizations
/24 → network 24 bits, host 8 bits → very small networks

You can **customize the network size** exactly according to the organization's need → no wasted addresses

What the subnet mask does
Subnet mask tells **which bits belong to the network** and **which belong to hosts**.

Formula:
Network Address = IP Address AND Subnet Mask
By changing the number of bits in the **network portion**, you can:

1. **Create more networks** (borrow host bits → more network bits)
2. **Allow more hosts per network** (use fewer network bits → more host bits)

Routers and devices use **IP AND subnet mask** to figure out the **network address**.

Subnetting is for internal advantages
Subnetting doesn't change your global allocation. It just helps you:

Organize your network → e.g., separate departments, floors, or servers
Use addresses efficiently → no wasted host addresses
Control traffic and security → you can isolate subnets

Outside your organization: routers and the Internet only see your **network ID as a whole** (your assigned block or public IP if NATed).

When your device sends a request
Your device (phone, computer) has:
IP address (e.g., 192.168.1.5)
Subnet mask (e.g., 255.255.255.0)

Inside your local network (LAN):
Subnet mask is used by your router to figure out **which network your device belongs to** AND operation: IP & subnet mask = network address
This helps the router **deliver packets to the correct internal device**

2. When the packet goes outside (Internet)
Once your router forwards the packet to the Internet:

Your **private IP is replaced by the router's public IP** (NAT)
Only the **public IP** is used for routing
Subnet mask is not sent across the Internet
Routers on the Internet only use the **network ID (from public IP)** to forward packets

In other words: **subnet mask is only relevant inside your own network**.

Classful IP: Fixed network and host boundaries based on classes (A, B, C).
Classless IP (CIDR): Flexible network boundary; NetID can be any number of bits.

Classful vs Classless IP

Class	NetID	HostD	Example
A	1 octet	3 octets	10.0.0.5 → NetID: 10, HostID: 0.0.5
B	2 octets	2 octets	172.16.5.10 → NetID: 172.16, HostID:5.10
C	3 octets	1 octet	192.168.1.10 → NetID: 192.168.1, HostID:10

Classless Inter-Domain Routing (CIDR): NetID can be any number of bits (e.g., 192.168.1.10/26 → first 26 bits NetID, last 6 bits HostID).

Math:
Given network: **172.16.0.0/16**
We need **at least 8 subnets**.

Step 1: How many bits to borrow?
Original prefix = /16
That means:
First 16 bits → network
Last 16 bits → host

To create subnets, we **borrow bits from the left side of the host portion** (most significant host bits).
We need at least 8 subnets.
Use:
$2^n \geq$ required subnets
$2^3 = 8$
So, we must borrow **3 bits from the left side of the host part**.

Step 2: New Subnet Mask
Original mask = /16
After borrowing 3 bits:
New prefix = /16 + 3 = **/19**
To calculate the dotted mask:

1. Write original mask in binary:

255.255.0.0
11111111.11111111.00000000.00000000

2. Borrow 3 bits from the left side of the host portion
   That means we **set those 3 borrowed bits to 1**.

So third octet becomes:
11100000
(First 3 bits are 1 because we borrowed them)

3. Convert 11100000 to decimal:

128 + 64 + 32 = 224
So new subnet mask:
255.255.224.0

Step 3: Hosts per Subnet
Original host bits = 16
Borrowed 3 bits → remaining host bits after borrowing = 13
Total IPs per subnet:
$2^{13} = 8192$
Usable hosts:
8192 − 2 = **8190 hosts**

Final Answer
Network: 172.16.0.0/16
Bits borrowed (from left of host portion): 3
New subnet mask: **255.255.224.0 (/19)**
Number of subnets: 8
Usable hosts per subnet: **8190**

Borrowing 3 bits does two important things:
1. It creates 8 subnets
2. It identifies which subnet a particular IP belongs to

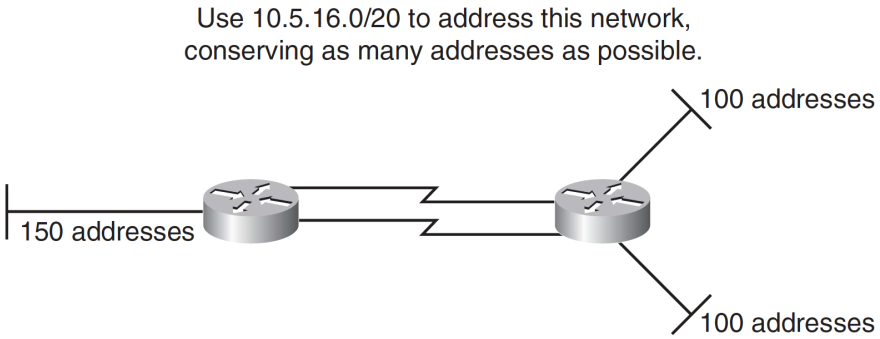
Subnet Bits → Subnet Number → Network Address
000 → 1st Subnet → 172.16.0.0/19
001 → 2nd Subnet → 172.16.32.0/19
010 → 3rd Subnet → 172.16.64.0/19
011 → 4th Subnet → 172.16.96.0/19
100 → 5th Subnet → 172.16.128.0/19
101 → 6th Subnet → 172.16.160.0/19
110 → 7th Subnet → 172.16.192.0/19
111 → 8th Subnet → 172.16.224.0/19

Applying a subnet mask (AND operation) to an IP address **finds the network address of the subnet where the IP belongs to**.

Slide: Wk5-3-Subnetting-VLSM

Variable-Length Subnet Masks (VLSM)
VLSM allows you to use different subnet sizes inside the same network, based on actual host requirements. This avoids wasting IP addresses compared to fixed-length subnetting.

Important Rule
If you use any addresses from a subnet, that subnet must not be further subnetted. So planning must be done from largest requirement to smallest.



Given Network
Base network: 10.5.16.0/20
A /20 network has:
32 − 20 = 12 host bits
Total addresses = 2^{12} = 4096

Step 1: Left LAN (150 hosts)

Required hosts = 150
Next power of 2 ≥ 150 → 256
256 = 2^8 → 8 host bits needed
Subnet mask = /24 (32 − 8 = 24)

So:

Left LAN uses a **/24 subnet**
Hosts available = 254

Step 2: Other Two LANs (100 hosts each)
For each LAN:

Required hosts = 100
Next power of 2 ≥ 100 → 128
128 = 2^7 → 7 host bits needed
Subnet mask = /25 (32 − 7 = 25)

So:

Each LAN uses a **/25 subnet**
Hosts available = 126 per LAN

Step 3: WAN Links

WAN links typically need 2 usable host addresses.
Required hosts = 2
2^2 = 4 total addresses
Host bits needed = 2
Subnet mask = /30 (32 − 2 = 30)

Each WAN subnet:

Total addresses = 4
Usable hosts = 2

We need subnets for:

1. Left LAN → 150 hosts → /24
2. Two LANs → 100 hosts each → /25
3. WAN links → 2 hosts each → /30

Why /24 Subnets Are Possible Here

You said:
Because at most 8 host bits are needed, the 10.5.16.0/20 address can be further subnetted into sixteen /24 subnets
Explanation:

/20 → /24 means borrowing **4 bits**
2^4 = **16 subnets**
Each /24 has **8 host bits** (256 addresses)

This works because:

The **largest LAN needs only 8 host bits**
So dividing the /20 into /24 blocks is valid

Key Exam Points

Always assign **largest subnet first**
Once a subnet is used, **do not subnet it further**
VLSM minimizes IP address wastage
/20 can be split into:

16 × /24 or mixed sizes using VLSM

In VLSM, the network is allocated by assigning the largest required subnet first, then progressively smaller subnets from the remaining address space, without further subnetting assigned blocks.

Different LANs need separate IP networks because routers only forward traffic between different networks, not within the same one.

The Core Rule of IP Networking
One IP subnet = one broadcast domain

Devices in the same subnet communicate directly
Devices in different subnets need a router (Layer 3 device)

So if you have **multiple LANs**, each LAN **must have its own subnet**.

Why Routers Don't Forward Traffic Inside the Same Subnet
Routers operate at **Layer 3 (Network Layer)**.
Router logic:

If source and destination are in the **same network → do nothing**
If they are in **different networks → route the packet**

Why?

Routing is meant to **connect networks**
Switches already handle same-network traffic efficiently

So: **Routers forward traffic between networks, not within them.**

An IP address identifies one network interface (one location) on a device.
Not: the whole router or the whole LAN

But: **a single interface / connection point**

Each physical port on a router is a separate interface. Each interface can be assigned:

Its own IP address
Its own subnet/network

Math:

VLSM Math: [VLSM Subnetting Tutorial]

Step-by-Step Process to solve a VLSM subnetting math problem

Step 1: List Requirements

1. Note the main network and its prefix (e.g., 10.5.16.0/20)
2. List all subnets/departments and the number of hosts needed

Step 2: Sort Subnets by Size

1. Arrange the subnets from **largest host requirement to smallest**
2. This avoids wasting IP addresses

Step 3: Allocate First Subnets

1. Start with the **largest subnet first**
2. Assign the network address from the starting main network
3. Set all **host bits to 0** → network address
4. Set all **host bits to 1** → broadcast address

Step 4: Calculate Host Bits for Each Subnet

1. Use formula: 2^h − 2 ≥ required hosts
2. Solve for **h (number of host bits)**

Step 5: Calculated Next Network Address:

1. You know the **broadcast address** of that subnet.
2. The **next subnet** always starts at **the next IP after the broadcast address**, because the next ip is the first available IP is not yet used.

Step 6: Calculate subnet mask:

1. We have the value of h from step 4, so subnet is **32 − h**
2. Or simply set **Network bits = 1**, **host bits = 0** in binary
3. Convert each 8-bit group (octet) to decimal → that's the subnet mask

Step 7: Calculated Broadcast Address:

1. Take the network address of the subnet
2. Identify the host bits — these are the bits that are left for hosts after the subnet mask.
3. Set all host bits to 1 → that gives the broadcast address in binary.
4. Convert each octet to decimal: the host bits might affect one or more octets.
5. Add the decimal value of host bits to the respective octet(s) of the network address to get the broadcast.

Step 8: Calculate Host Range

1. First usable host = network + 1
2. Last usable host = broadcast − 1

Step 9: Move to Next Subnet

1. Take the **next available IP** after the previous broadcast
2. Repeat **Steps 4–7** for each remaining subnet

Questions:

A company is assigned the IP address block 10.5.16.0/20.
The company needs to create subnets using VLSM (Variable Length Subnet Masking) for the following number of connected hosts: **700** hosts **400** hosts **150** hosts **150** hosts **100** hosts **2** hosts **2** hosts.
For each required subnet, determine: Network Address Subnet Mask (in decimal and CIDR notation) Broadcast Address

Answer:

Main Network: **10.5.16.0/20**
Subnet Mask: **255.255.240.0**
Range: 10.5.16.0 – 10.5.31.255

for 700 → need 2^10 − 2 ≥ 700
10.5.00010000.00000000
Network Address: 10.5.16.0
Subnet Mask: 255.255.252.0/22
Broadcast Address: 10.5.19.255
for 400 → need 2^9 − 2 ≥ 400
10.5.00010100.00000000
Network Address: 10.5.20.0
Subnet Mask: 255.255.254.0/23
Broadcast Address: 10.5.21.255

To write this binary notation it will be the next bit of previous broadcast address
Always consider its own binary notation for subnet and broadcast calculations.

for 150 → need 2^8 − 2 = 254 → /24
10.5.00010110.00000000
Network Address: 10.5.22.0
Subnet Mask: 255.255.255.0/24
Broadcast Address: 10.5.22.255

for 150
10.5.00010111.00000000
Network Address: 10.5.23.0
Subnet Mask: 255.255.255.0/24
Broadcast Address: 10.5.23.255

for 100 → need 2^7 − 2 = 126 → /25
10.5.00011000.00000000
Network Address: 10.5.24.0
Subnet Mask: 255.255.255.128/25
Broadcast Address: 10.5.24.127

for 2 → need 2^2 − 2 = 2 → /30
10.5.00011000.10000000
Network Address: 10.5.24.128
Subnet Mask: 255.255.255.252/30
Broadcast Address: 10.5.24.131

for 2
10.5.00011000.10000100
Network Address: 10.5.24.132
Subnet Mask: 255.255.255.252/30
Broadcast Address: 10.5.24.135

Short Method:

1. Binary notation and network address will be same.
2. Mark host and network bit required for subnet using 2^h
3. Calculate subnet mask making all network bit set.
4. Calculate broadcast address after setting all host bit 1 from notation.

N.B: Broadcas address will be Network address + value of set host bit.

Slide: Wk5-4-Routing

Routing (Network Layer OSI -3)

Network routing is the process of selecting the most appropriate path for data packets to travel from a source device to a destination across a network.

It is performed by routers using routing tables and operates at the Network Layer (Layer 3) of the OSI model. It is a network-wide process.

Forwarding: The process of moving a packet from a router's input interface to the correct output interface. It is a local router operation.

Routing Algorithm:
The part of the network layer responsible for deciding **which output line a packet should use** at each router.

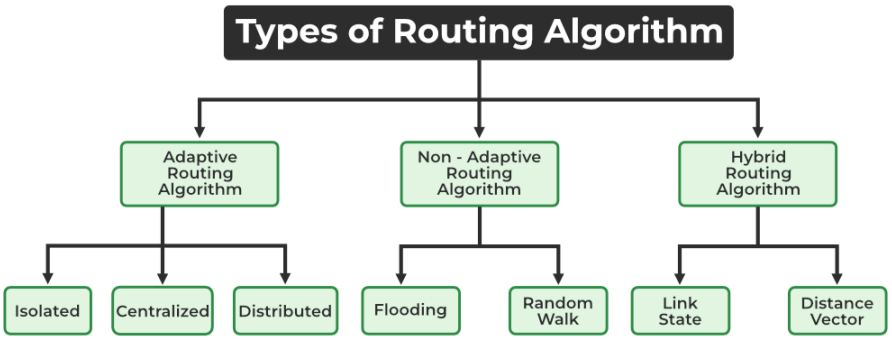
Components of Routing

1. Routers
2. Routing tables
3. Routing algorithms
4. Routing protocols
5. Metrics (hop count, cost, bandwidth, delay)

Routing Metrics:
Routing metrics are values used by routing protocols to select the best path for data transmission.

Common routing metrics include:

1. **Hop Count:** Number of routers a packet passes through.
2. **Bandwidth:** Data transmission capacity of a link.
3. **Delay:** Time taken for a packet to travel from source to destination.
4. **Load:** Current traffic level on a link.
5. **Reliability:** Stability and error rate of a link.
6. **Cost:** Administratively assigned value representing path preference.



Types of Routing:

1. **Static Routing:** Routes are manually configured by the network administrator.
2. **Dynamic Routing:** Routes are automatically learned and updated using routing protocols.
3. **Default Routing:** A predefined route used when no specific route to a destination is found.

Static Routing

Static routing is when routes are **manually configured** by a network administrator and **do not update automatically**.

Advantages:

1. Simple to configure
2. Low overhead (no routing protocol traffic)

Disadvantages:

1. Not scalable for large networks
2. No automatic fault tolerance (does not adjust if a link fails)

Dynamic Routing

Dynamic routing is when routers **automatically learn and update routes** by exchanging routing information with other routers.

Advantages:

1. Scalable for large networks
2. Adapts automatically to network changes

Disadvantages:

1. More overhead (routing updates consume resources)
2. More complex to configure and manage

Routing Algorithms

1. **Distance Vector Algorithm:** Routers share routing tables with neighbors and choose paths based on distance (e.g., hop count).
2. **Link State Algorithm:** Routers build a complete map of the network and calculate the shortest path.
3. **Path Vector Algorithm:** Routers advertise the entire path to a destination to avoid routing loops (used in inter-domain routing).

Distance Vector Routing

1. Each router knows the **distance to its neighbors**.
2. Routers **periodically share routing tables** with directly connected neighbors.
3. Uses the **Bellman–Ford algorithm** to calculate shortest paths.
4. Example protocol: **Routing Information Protocol (RIP)**.

Problems in Distance Vector Routing

1. **Routing Loops:**
   Routers may continuously forward packets in a circular path.
2. **Count-to-Infinity Problem:**
   After a link failure, routers slowly increase the metric value step by step until it reaches infinity (very slow convergence).

Solutions:

1. **Split Horizon:**
   A router does not advertise a route back on the interface from which it was learned.
2. **Poison Reverse (Route Poisoning):**
   A failed route is advertised with an infinite metric (e.g., 16 in RIP) to indicate it is unreachable.
3. **Hold-Down Timers:**
   A router waits for a specific time before accepting new updates for a recently failed route to prevent incorrect information.

Link State Routing

1. Each router builds a complete map of the network topology.
2. Routers flood **link-state** information to all other routers.
3. Uses Dijkstra's shortest path algorithm to compute the best path.
4. Example protocols:
   5. Open Shortest Path First (OSPF)
   6. Intermediate System to Intermediate System (IS-IS)

Distance Vector vs Link State

Feature	Distance Vector	Link State
Knowledge	Knows neighbors only	Knows full topology
Updates	Periodic	Event-driven
Convergence	Slow	Fast
Complexity	Low	High

Path Vector Routing

1. Used for **inter-domain routing** between autonomous systems.
2. Maintains the **full path information** for each route to prevent routing loops.
3. Example protocol: **Border Gateway Protocol (BGP)**.

Route Summarization:
Route Summarization is combining multiple smaller networks into a single larger network in the routing table to reduce its size and simplify routing.

1. **Without summarization:** Router lists each network separately (192.168.1.0/24, 192.168.2.0/24...).
2. **With summarization:** Router combines them into one bigger network (192.168.0.0/22) to make the routing table smaller.

Interior vs Exterior Routing

Interior Gateway Protocols (IGP):

a. Used **within an organization** (autonomous system)
b. Examples: **Routing Information Protocol (RIP)**, **Open Shortest Path First (OSPF)**, **Intermediate System to Intermediate System (IS-IS)**

Exterior Gateway Protocols (EGP):

c. Used **between organizations** (between autonomous systems)
d. Example: **Border Gateway Protocol (BGP)**

Real-World Routing Protocols

1. **Routing Information Protocol (RIP):** Simple, suitable for small networks.
2. **Open Shortest Path First (OSPF):** Used in large enterprise networks.
3. **Border Gateway Protocol (BGP):** Used for Internet backbone routing.

Routing Table Structure
A routing table typically contains:

1. **Destination Network:** The network or subnet the packet is destined for.
2. **Next Hop:** The IP address of the next router to forward the packet to.
3. **Metric:** Value used to determine the best path (e.g., hop count, cost).
4. **Interface:** The local network interface through which the packet should be sent.

Challenges in Routing

1. **Scalability:** Handling growth in network size and routing information.
2. **Convergence Time:** Time taken for all routers to have consistent routing information.
3. **Loop Prevention:** Avoiding routing loops that can cause packet loss or delays.
4. **Security:** Protecting routing information from attacks or tampering.

Flooding
Flooding is a method where every incoming packet is sent out on all outgoing links except the one it arrived on. It's simple and doesn't need a routing table, but it can create many duplicate packets.

Random Walk
Random Walk, on the other hand, sends the packet randomly through one of the outgoing links (again, not the incoming link). It can be seen as a mix between adaptive and non-adaptive routing techniques.

Optimality Principle
If a path from one router to another is the shortest (optimal), then any part of that path is also the shortest path between the intermediate routers on that path.

1. If a path from Router I → Router K is the shortest (optimal),
2. Then any part of that path is also the shortest between its endpoints.

In other words:
You can split a shortest path into smaller segments, and each segment is also the shortest path.

Example

1. Optimal path: I → A → J → K
2. Then the path from J → K is also the shortest path: J → K
3. There are no unnecessary detours in any segment.

Why It Matters

1. Makes Distance Vector and Shortest Path algorithms work correctly.
2. Ensures routing tables are consistent.
3. Helps routers always pick the shortest route.

ARP and RARP

ARP (Address Resolution Protocol) is a network protocol used to find the hardware (MAC) address of a device when its IP address is known.

Address Resolution Protocol (ARP)

1. **Purpose:** Map a 32-bit IP address to a 48-bit MAC address
2. **Used by:** Hosts to find hardware addresses for local delivery
3. **Example:**
   IP: 192.168.1.5 → ARP → MAC: 00:1A:2B:3C:4D:5E

RARP (Reverse Address Resolution Protocol) is a network protocol used to find the IP address of a device when its MAC address is known.

Reverse Address Resolution Protocol (RARP)

1. **Purpose:** Map a 48-bit MAC address to a 32-bit IP address
2. **Used by:** Diskless machines or devices without an IP to discover their IP
3. **Example:**
   MAC: 00:1A:2B:3C:4D:5E → RARP → IP: 192.168.1.5

Slide: Wk6-1-Transport Layer

The Transport Layer is responsible for process-to-process delivery of data between hosts. It ensures data from one host's application reaches the correct application on another host.

Various Responsibilities of the Transport Layer

1. Process to process delivery
2. End-to-end connection between hosts
3. Multiplexing and demultiplexing
4. Congestion control
5. Data integrity and error correction
6. Flow control

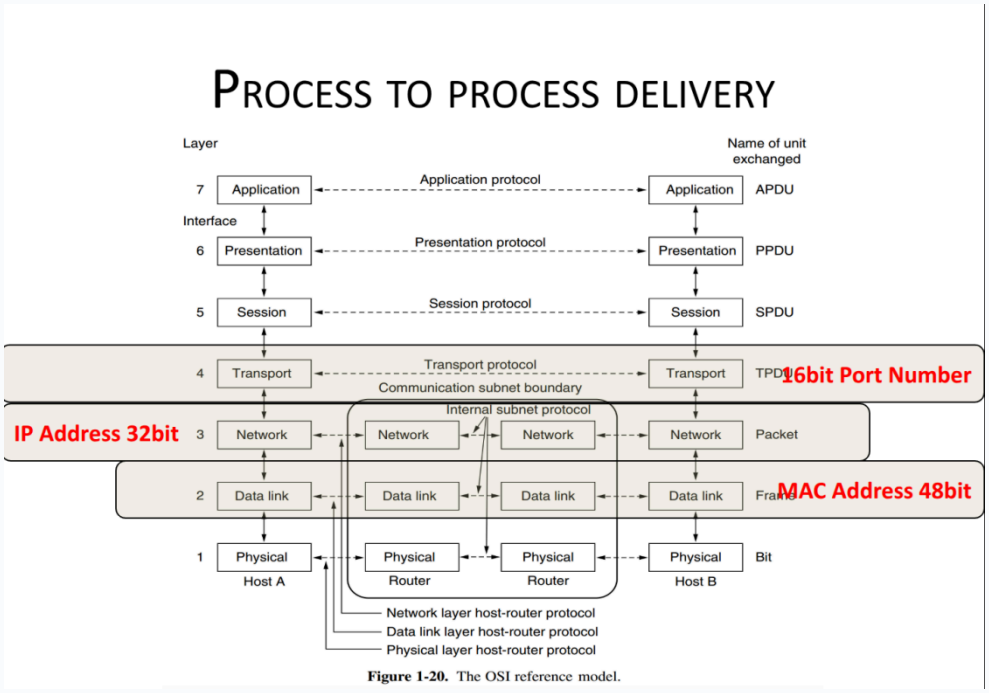


Figure 1-20. The OSI reference model.

Process to Process Delivery

1. The transport layer delivers data **from one application process to another**, not just from one host to another.
2. This is achieved using **port numbers**, which uniquely identify application processes.
3. Example: Web browser (port 80/443), Email service (port 25).

Port Numbers and Sockets

- Port number identifies an application
- Socket = IP address + Port number
- Examples:
  - HTTP: Port 80
  - HTTPS: Port 443
  - FTP: Port 21
- Enables multiple applications to use the network simultaneously

End-to-End Connection Between Hosts

1. Transport layer establishes logical communication between **source and destination hosts**.
2. Two types of connections exist:
   3. Connection-oriented
   4. Connectionless

Types of Transport Layer Protocols

Connection-Oriented Protocol

1. Devices must **open a connection** before transmitting data and **close it gracefully** after transmission.
2. Data is delivered reliably and in order.
3. Example: **TCP (Transmission Control Protocol)**

Connectionless Protocol

1. No prior connection setup.
2. Data is sent independently.
3. Faster but unreliable.
4. Example: **UDP (User Datagram Protocol)**

Transmission Control Protocol (TCP)

1. **Connection-Oriented:** Devices must **establish a connection** before data transfer and **close it gracefully** afterward.
2. **Reliable Delivery:** Ensures all data reaches the destination correctly.
3. **Error Checking:** Provides **flow control**, **acknowledgments**, and retransmission of lost packets.
4. **Sequencing:** Maintains the correct order of data.
5. **Multiplexing/Demultiplexing:** Uses **TCP port numbers** to support multiple applications on the same host.
6. **Performance:** Slower than UDP due to extensive error checking and reliability mechanisms.

Features of TCP

1. Stream data transfer
2. Reliability
3. Flow control
4. Multiplexing
5. Logical connections
6. Full duplex communication

User Datagram Protocol (UDP)

UDP is a **datagram-oriented, connectionless protocol**. No connection setup, maintenance, or termination.

[Transport Layer Protocol Details Article]

User Datagram Protocol (UDP)

1. **Connectionless:** No need to establish or terminate a connection; no three-way handshake.
2. **Efficient:** Suitable for **broadcast or multicast** transmissions.
3. **Basic Error Checking:** Uses **checksums** only.
4. **No Sequencing:** Data may arrive out of order.
5. **Unreliable Delivery:** Does **not guarantee** that data reaches the destination.
6. **Fast and Simple:** Less overhead and more efficient than TCP.
7. **Multiplexing/Demultiplexing:** Uses **UDP port numbers** for multiple applications.
8. **No Retransmission:** Lost packets are not resent.

Situations Where UDP Is Preferred

1. SNMP (Simple Network Management Protocol)
2. DNS (Domain Name System)
3. Internet radio / streaming
4. LAN requests
5. Router update exchanges

Differences Between TCP and UDP

Basis	TCP	UDP
Definition	Establishes a virtual circuit before data transmission	Sends data directly without checking receiver readiness
Connection Type	Connection-oriented	Connectionless
Speed	Slow	High
Reliability	Reliable	Unreliable
Header Size	20 bytes	8 bytes
Acknowledgement	Uses ACK and retransmits lost packets	No ACK and no retransmission

Multiplexing and Demultiplexing

1. **Multiplexing** allows multiple applications on a host to send data simultaneously.
2. **Demultiplexing** delivers received data to the correct application.
3. Achieved using **port numbers**.

Congestion Control

Congestion means overcrowding in the network.

1. Occurs when too many sources send data simultaneously.
2. Router buffers overflow, causing packet loss.
3. Retransmission of lost packets increases congestion further.
4. TCP uses congestion control mechanisms to manage this issue.

Data Integrity and Error Correction

1. Transport layer checks errors using **checksums**.
2. Ensures data is not corrupted during transmission.
3. Uses **ACK (Acknowledgement)** and **NACK (Negative Acknowledgement)**.
4. Confirms correct and complete delivery of data.

Flow Control

1. Manages data transmission between a **fast sender** and a **slow receiver**.
2. Prevents receiver buffer overflow.
3. TCP uses flow control mechanisms like **sliding window**.

TCP Connection Establishment and Release

TCP Connection Establishment
Three-Way Handshake
TCP establishes a reliable connection using a **three-way handshake** mechanism before data transmission begins.

Steps of Three-Way Handshake

1. **SYN (Synchronize)**
    a. The client sends a **SYN** segment to the server.
    b. Indicates a request to establish a connection.
    c. Includes the client's initial sequence number.
2. **SYN + ACK (Synchronize + Acknowledge)**
    a. The server responds with **SYN + ACK**.
    b. ACK acknowledges the client's SYN.
    c. SYN contains the server's initial sequence number.
3. **ACK (Acknowledge)**
    4. The client sends an **ACK** back to the server.
    5. Acknowledges the server's SYN.
    6. Connection is successfully established.
    After these three steps, both sides are ready to transmit data.

Purpose of Three-Way Handshake

1. Ensures both sender and receiver are ready.
2. Synchronizes sequence numbers.
3. Prevents duplicate or delayed connections.
4. Establishes a reliable communication channel.

TCP Connection Release (Termination)

TCP connection termination is **graceful** and normally uses a **four-step process**.

Steps of TCP Connection Release
FIN (Finish): The host that wants to close the connection sends a **FIN** segment. Indicates no more data to send.

ACK: The receiving host sends an **ACK** to acknowledge the FIN.
Connection becomes half-closed.

FIN: The receiving host again sends its own **FIN** when it is ready to close.

ACK: The initiating host sends the final **ACK**.
Connection is completely terminated.

Key Points

TCP uses **three-way handshake** for connection establishment.
TCP uses **four-way termination** for connection release.
Ensures reliable, ordered, and error-free communication.
Prevents data loss and premature connection closure.

Slide: Wk7-1- Presentation Layer

The **Presentation Layer** is responsible for **syntax and semantics of the information exchanged between two systems**. It ensures that data sent by the application layer of one system can be understood by the application layer of another system.

Specific Responsibilities of Presentation Layer

1. Data representation
2. Data compression
3. Encryption

Data Representation

1. Different computers use different internal representations for data such as characters, integers, and floating-point numbers.
2. When two **dissimilar computers** exchange data (for example, e-mail), they must agree on a **common data format**.
3. **The presentation layer:**
    4. Translates data from the sender's format to a standard format.
    5. Converts received data from the standard format to the receiver's local format.
6. This ensures **interoperability between heterogeneous systems**.

Data Compression

1. Compression is used to:
a. Maximize the use of available **network bandwidth**
b. Reduce **storage space** when saving data
2. Compression reduces redundancy in data before transmission and storage.

Lossless Compression

1. Lossless compression ensures that **decompressed data is exactly the same as the original data**.
2. There is **no loss of information**.
3. Used for data that cannot tolerate errors, such as:
    a. Executable programs, Text files, Numeric and database data
4. Generally provides **less compression ratio** than lossy compression.
5. Requires **more processing power**.

Lossless Compression Algorithms

Run Length Encoding (RLE)

1. Replaces consecutive occurrences of a symbol with:
    a. One copy of the symbol
    b. A count of its repetitions
    Example: AAABBCDDDD → 3A2B1C4D
2. Effective when data has many repeated symbols.
3. Used in fax machines.
4. May increase file size if data has little repetition.

Differential Pulse Code Modulation (DPCM)

1. Uses a **reference symbol**.
2. Stores the **difference** between each symbol and the reference symbol.
3. Example:
    Using A as reference:
    AAABBCDDDD → A0001123333
4. Reduces redundancy when consecutive symbols are similar.

Dictionary-Based Encoding

1. Uses a dictionary of commonly occurring strings.
2. One popular algorithm is **Lempel–Ziv (LZ)**.
3. Also known as **substitution coding**.
4. Frequently occurring strings are replaced by dictionary indexes.
5. Example:
    If the word "compression" has index 4978, every occurrence is replaced with 4978.

Lossy Compression

1. Lossy compression removes some data permanently.
2. Decompressed data is **not identical** to the original.
3. Used when **perfect accuracy is not required**.
4. Commonly applied to: Audio, Image, Video data

Audio Compression Techniques

1. Predictive Encoding

1. Encodes the **difference between successive samples** instead of full values.
2. Commonly used for speech compression.
3. Standards include:
    a. GSM (13 kbps)
    b. G.729 (8 kbps)
    c. G.723.3 (6.4 or 5.3 kbps)

2. Perceptual Encoding

1. Based on **psychoacoustics**, the study of human sound perception.
2. Exploits limitations of the human auditory system.
3. Removes sounds that are **masked by louder sounds**.
4. Example: Loud hammer noise can mask flute music.
5. The compressed signal may look different on instruments but sounds the same to humans.

What is SSL?

Definition:

SSL (Secure Sockets Layer) is a cryptographic protocol that secures communication over the Internet. It makes sure that any data sent between a user's browser and a web server is encrypted and protected from eavesdroppers or hackers.

Purpose of SSL:

1. **Encryption:** Converts readable data into a secure, scrambled format so outsiders cannot read it.
2. **Authentication:** Confirms the identity of the web server to ensure you are connecting to the right site.
3. **Data Integrity:** Ensures the data is not altered or tampered with during transmission.

Step 1: Browser Connects to the Server

When you type a website address starting with **https://**, your browser sends a request to the server asking to establish a **secure connection**.

Step 2: Server Sends Its SSL Certificate

The server responds by sending its **SSL certificate**, which proves its identity to the browser. This certificate contains:

1. The server's public key
2. The domain name
3. The certificate authority (CA) that issued it

The browser checks the certificate to make sure it's valid and issued by a trusted authority.

Step 3: Browser Verifies the Certificate

Your browser checks the server's SSL certificate to make sure:

1. The certificate is **valid** and not expired.
2. It matches the **domain name** you are visiting.
3. It is issued by a **trusted Certificate Authority (CA)**.

If all checks pass, the browser trusts the server.

Step 4: Handshake and Encryption
Once the certificate is verified, the browser and server perform a **TLS/SSL handshake**:

1. They exchange **public and private keys** to securely agree on a shared secret key.
2. This shared key is then used for **fast, encrypted communication** between the browser and server.

Wk8-1.Application Layer-DNS-DHCP-EMAIL

Dynamic Host Configuration Protocol (DHCP)
Why an IP address matters:
Every device (computer, phone, printer, etc.) on a network needs a unique IP address to communicate. Without one, it can't send or receive data on the network.

Manual IP configuration problems:

1. **Time-consuming:** Imagine assigning IPs one by one for hundreds of devices.
2. **Error-prone:** Mistakes like duplicate IPs can cause network conflicts.
3. **Difficult to manage at scale:** Keeping track of assigned IPs manually gets messy fast.

How DHCP helps:

1. DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices when they connect to the network.
2. It ensures each device gets a unique IP without manual input.
3. It can also assign other network info automatically, like subnet mask, default gateway, and DNS servers.

You can think of DHCP as a "network receptionist" that hands out addresses to devices so they can start talking to each other immediately, without the administrator having to assign them manually.

DHCP is: DHCP stands for **Dynamic Host Configuration Protocol**. It works at the **application layer** of the network

What it assigns automatically:

IP address: So the device can communicate on the network.
Subnet mask: Defines which part of the IP is the network and which part is the host.
Default gateway: Lets the device send traffic outside the local network.
DNS server: Helps the device translate domain names (like google.com) into IP addresses.

Client-server model:
The **DHCP client** is the device asking for network info (like your computer).
The **DHCP server** is the system that gives out IP addresses and other settings.

So basically, DHCP saves a lot of time and prevents errors by letting devices automatically get all the information they need to join a network.

DHCP Components

DHCP Client

1. This is the device that needs an IP address and other network info (computer, phone, printer, etc.).
2. It sends requests to the DHCP server when it connects to the network.

DHCP Server

1. This is the system that **owns the pool of IP addresses** and other network settings.
2. It **allocates IPs** and provides the subnet mask, gateway, DNS, etc.
3. Can be a dedicated server or built into a router.

DHCP Relay Agent (optional)

1. Some networks have **multiple subnets**, and the DHCP server might be on a different subnet than the client.
2. The relay agent **forwards DHCP requests** from clients to the server and relays the replies back.
3. Usually a router can act as a DHCP relay.

DHCP Uses which Layer?

1. **Layer:** Application layer protocol
2. **Transport:** Uses UDP
a. **Server port:** 67
b. **Client port:** 68
3. **Works over:** TCP/IP stack to deliver IP configuration quickly without needing a connection

UDP is used because it's fast and lightweight, perfect for automatic IP assignment.

DHCP Address Allocation Methods

Dynamic Allocation: Temporary IP assigned for a limited time (lease).
Automatic Allocation: Permanent IP assigned from a pool.
Manual Allocation (Reservation): Fixed IP assigned based on device's MAC address.

DHCP Lease Concept

1. IP address is assigned for a lease time
2. Client must renew before expiration
3. Helps reuse IP addresses efficiently

DHCP Message Exchange (DORA)

1. **Discover** – Client broadcasts request
2. **Offer** – Server offers IP address
3. **Request** – Client requests the offered IP
4. **Acknowledge** – Server confirms assignment
   (*This process is called* DORA)

Message Flow
Step-by-Step Communication
Client → **Server** DHCP Client broadcasts Discover messages
Server → **Client** DHCP Server unicasts Offer messages
Client → **Server** DHCP Client responds with Request message
Server → **Client** DHCP Server completes the process with acknowledgment message

Advantages & Limitations

Advantages
Automatic IP management
Reduces configuration errors
Easy network scalability
Centralized control

Limitations
DHCP server failure affects clients
Not suitable for devices needing fixed IPs
Security risks if rogue DHCP server exists

DHCP vs Static IP

DHCP	Static IP
Automatic	Manual
Temporary	Permanent
Easy to manage	More control
Used in large networks	Used for servers

Common Uses of DHCP

- Home networks
- Office LANs
- University campuses
- Wi-Fi networks
- Mobile and IoT devices

Domain Name System (DNS)

Humans remember names, computers use IP addresses

Example:

Domain name: www.google.com
IP address: 142.250.190.14

DNS translates domain names to IP addresses

What is DNS

DNS is a distributed naming system
Maps: Domain names → IP addresses
Works as a hierarchical and decentralized system
Part of the Application Layer

Ports Used

Uses UDP port 53 for normal queries
TCP port 53 for zone transfer and large responses
Client side Random Port 1024+

Hierarchical Tree Structure

Root (.)
Top-Level Domain (TLD)
com, .org, .edu, .bd
Second-Level Domain
google.com
Subdomain
www or mail
This is often the leftmost label in a fully qualified domain name (FQDN)

DNS Namespace Structure

The specific name assigned to a device or service

Example: mail.google.com

Limits

1. **Total Character Limit:** The entire FQDN cannot exceed 253 characters
2. **Label Limit:** Each individual label between dots can have a maximum of 63 characters
3. **Depth Limit:** While 127 levels are theoretically possible, most browsers and networks operate within 3 to 5 levels for practical efficiency

DNS Components

1. DNS Resolver
    a. Client-side (browser or OS)
2. Root Name Server
    a. Knows TLD servers
3. TLD Name Server
    a. Knows authoritative servers
4. Authoritative Name Server
    a. Stores actual DNS records

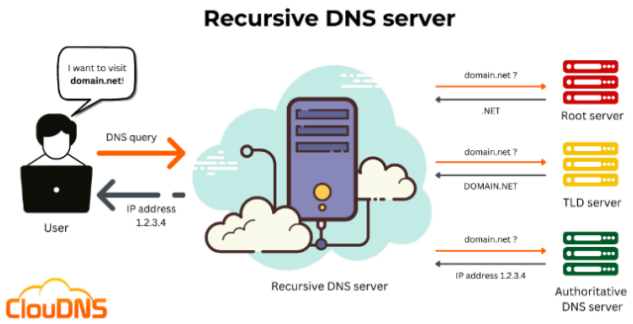
DNS Servers Involved in Loading a Webpage

1. DNS Recursor
2. Root Nameserver
3. TLD Nameserver
4. Authoritative Nameserver

DNS Recursor

The recursor can be thought of as a librarian who is asked to go find a particular book somewhere in a library

A server designed to receive queries from client machines through applications such as web browsers



TLD Name Server

A TLD (Top-Level Domain) nameserver is responsible for directing queries to the appropriate authoritative nameserver for a domain

Management

Management of TLD nameservers is handled by the Internet Assigned Numbers Authority (IANA), which is a branch of ICANN

Types of TLDs

Generic Top-Level Domains (gTLDs)

These are domains that are not country specific

Examples:

.com
.org
.net
.edu
.gov

Country Code Top-Level Domains (ccTLDs)

These include domains specific to a country or state

Examples:

.bd
.uk
.us
.ru
.jp

Directory Server

1. A directory server is a specialized database for storing and managing structured information
2. Optimized for read-heavy operations
3. Commonly used for identity, authentication, and resource lookup

Examples

1. LDAP (Lightweight Directory Access Protocol)
2. Microsoft Active Directory
3. OpenLDAP
4. Apache Directory Server

RADIUS (*Remote Authentication Dial-In User Service*)

Directory Server in Campus Networks

1. Student and staff authentication
2. Access control to labs and Wi-Fi
3. Integration with learning management systems

LDAP Protocol Basics

1. Client-server model
2. Runs over TCP/IP
3. Default Ports: 389 (LDAP), 636 (LDAPS)
4. Operations Supported: bind, search, add, modify, Delete

FTP Server

1. An FTP server is a system that stores files and allows clients to upload or download files
2. Uses the File Transfer Protocol (FTP)
3. Works on a client-server model

Uses:

1. File sharing between computers
2. Uploading files to a web server
3. Downloading large files
4. Backup and data exchange

FTP Ports

1. Control connection: Port 21
2. Data connection: Port 20 (active mode)
3. Passive mode uses dynamic ports

Modes of Operation
Active Mode

1. Server initiates data connection
2. Client firewall issues possible

Passive Mode

1. Client initiates both connections
2. More firewall-friendly

FTP Commands

1. USER, PASS (authentication)
2. LIST (list files)
3. GET / RETR (download)
4. PUT / STOR (upload)
5. QUIT (close session)

FTP Server Software Examples

1. vsftpd (Linux)
2. FileZilla Server
3. ProFTPD
4. Microsoft IIS FTP Server

EMAIL SERVER

1. Simple Mail Transfer Protocol (SMTP)
2. To Send Email
3. Default Port: 25

What is a Proxy Server?

1. A proxy server is an intermediary system that acts on behalf of a client to request resources from another server
2. Client → Proxy → Destination Server → Proxy → Client
3. It operates mainly at the Application Layer (Layer 7)

Functions of a Proxy Server

1. Access control
2. Security enforcement
3. Caching
4. Anonymity
5. Traffic monitoring
6. Bandwidth optimization

Why Use a Proxy Server?

Basic Workflow

1. Client sends request to proxy
2. Proxy evaluates policy
3. Proxy forwards request to destination server
4. Server responds to proxy
5. Proxy sends response back to client
   Example: User → Proxy → Web Server

Types of Proxy Servers

1. Forward Proxy

1. Positioned between client and internet
2. Common in enterprise networks
3. Hides internal client IP addresses
4. Enforces browsing policies
5. Typical deployment in corporate LAN

2. Reverse Proxy

1. Positioned in front of web servers
2. Clients are unaware of backend servers
3. Used for: Load balancing, SSL termination, Security filtering

Common technologies:

1. NGINX
2. HAProxy
3. Apache HTTP Server

Transparent Proxy

Client is unaware of proxy presence
Often used by ISPs or institutions
Used for: Content filtering, Caching
No client-side configuration required

Proxy vs NAT

Feature	Proxy	NAT
Layer	Application	Network
Content Filtering	Yes	No
Caching	Yes	No
Protocol Awareness	Yes	Limited

Proxy and CDN Relationship

Content Delivery Networks use reverse proxy principles
Example: Cloudflare, Akamai Technologies
They cache content closer to users geographically

BOOTP

1. BOOTP (Bootstrap Protocol) is a network protocol used to automatically assign IP configuration information to a host during boot time
2. It was designed primarily for diskless workstations
3. Defined in RFC 951

BOOTP Characteristics

1. Application Layer protocol
2. Uses UDP
3. Client Port: 68
4. Server Port: 67
5. Transport Protocol: UDP (connectionless)

BOOTP Operation

Step 1: Client broadcasts request with MAC address
Step 2: Server matches MAC address in its configuration table
Step 3: Server sends:
IP address, Subnet mask, Default gateway, Boot file name, TFTP server address

BOOTP vs DHCP

Sr. No.	Key	BOOTP	DHCP
1	Definition	BOOTP stands for Bootstrap protocol.	DHCP stands for Dynamic Host Configuration Protocol.
2	Temporary IP Address	BOOTP has no support for temporary IP Addressing.	DHCP Server support for temporary IP Addressing but for limited period of time.
3	Client Support	BOOTP does not support DHCP Clients.	DHCP server supports BOOTP Clients.
4	Configuration Type	In BOOTP, configuration has to be done manually.	In DHCP, configuration is automatic.
5	Mobile Machine	Mobile machine are not supported.	Mobile machines are supported.
6	Error Probability	Configuration being manual often leads to errors.	Automatic configuration prevents any error to occur